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TurboAC Capture Examples

This section contains a program example for the TurboAC Capture instrument. This IG-XL workbook example shows all IG-XL data, VBT code, necessary for a complete running program.

The following topic is available:

- TurboAC Capture CW Down Conversion

For more information, see:

- TurboACCapture (VBT Syntax Reference)

- TurboACCapture Programming

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[TurboAC Capture Examples](#) : TurboAC Capture CW Down Conversion

TurboAC Capture CW Down Conversion

This workbook example includes the following program components:

- | | |
|---|-----------------|
| • | DUT Connections |
| • | DataTool Sheets |
| • | VBt Code |
| • | DSP Code |

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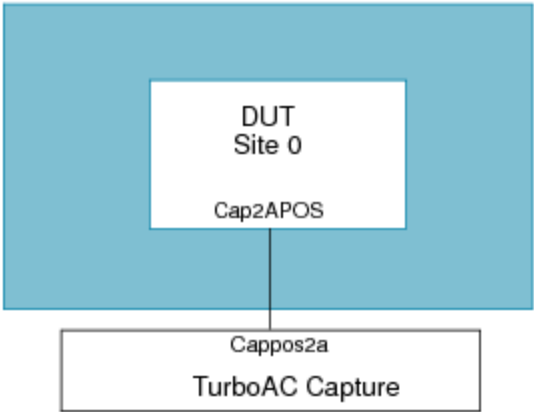
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TurboAC Capture Examples : TurboAC Capture CW Down Conversion : DUT Connections

DUT Connections



TurboAC Capture CW Down Conversion Connections

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TurboAC Capture Examples : TurboAC Capture CW Down Conversion : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet
- Mixed Signal Timing Sheet
- Wave Definitions Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	Cap2APos	Analog

Channel Map Sheet

Channel Map

DIB ID:

View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comm
Cap2APos		TurboACCapPo	11.cappos2a	

Test Instances Sheet

Test Instances				
Test Procedure				
Test Name	Type	Name	Called As	Catego
Capture_DownConversionVBT	VBT	CWtoDownConversionLoopback_VBT		

The test named Capture_DownConversionVBT has the following test-specific parameters and values, as shown in the instance editor:

Instance Editor [VBT Test 'CWtoDownConversionLoopback_VBT']

...

Lock

Down

Up

✓

Instance: Capture_DownConversionVBT_I

Parameters

Timing and Levels

Limits

	Name	Value	
☒	CapPin	Cap2APos	...
☐	CapVRange	1.28	...
☐	CapFs	49890377.5883069	...
☐	CapN	8192	...
☐	DownConvFreq	45000000	...
☐	IFreq	5000000	...

Flow Table Sheet

Flow Table

TERADYNE

Flow Domain:

Gate					Command		TName
Label	Enable	Job	Part	Env	Opcode	Parameter	
					Test	Capture_DownConversion	VBT_I
					Use-Limit	Capture_DownConversion	Freq
					Use-Limit	Capture_DownConversion	Amp
					Use-Limit	Capture_DownConversion	Offset
					Use-Limit	Capture_DownConversion	SNR
					Use-Limit	Capture_DownConversion	THD
					Use-Limit	Capture_DownConversion	SINAD
					Use-Limit	Capture_DownConversion	SFDR
					Use-Limit	Capture_DownConversion	IMD3A
					Use-Limit	Capture_DownConversion	IMD3B

Mixed Signal Timing Sheet

Mixed Signal Timing

TERADYNE

Set Name	Subset	Resource		Clocking					Instrument	Waveform		MSW
		Type	ID	Fs	N	Fr	M	USR	Data	Definition	Filter	Settings
Audio1KHzLoopback	Subset1	BBACSource	src1	20000	20	1000	1	1		aSine		loc=0,mnrelative
Audio1KHzLoopback	Subset1	BBACCapture	cap1	59904.936	32768	1000	547	1				

Wave Definitions Sheet

Wave Definitions

TERADYNE

WaveDefName	WaveDefType	WaveDef Component	Repeat Count	Relative Period	Relative Amplitude	Relative Offset	Primitive Parameters
aSine	Primitive	Sine	1	1.E+00	1.E+00	0	phase=0.0

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TurboACCapExamples.5.5

	
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TurboAC Capture Examples : TurboAC Capture CW Down Conversion : VBT Code

VBT Code

Public Function CWtoDownConversionLoopback_VBT(CapPin As PinList, CapVRange As Double, CapFs As Double, _

CapN As Double, DownConvFreq As Double, IFreq As Double, _

Optional TNames_ As String = _

"Freq|Amp|Offset|SNR|THD|SINAD|SFDR") As Long

Dim CapWave As New DSPWave

Dim Freq As New SiteDouble, Amp As New SiteDouble, Offset As New _
SiteDouble

Dim SNR As New SiteDouble, THD As New SiteDouble, _

SINAD As New SiteDouble, SFDR As New SiteDouble

Dim CapBin As New SiteDouble

TheHdw.TurboACCapture(CapPin).Connect tlTurboACCaptureFromPOS Or _
tlTurboACCaptureFromNEG, tlTurboACCaptureToCrossChannelLoopBack

' Set up the Turbo AC Capture Signal

With TheHdw.TurboACCapture(CapPin)

.Signals.Add "CapSig"

.Signals("CapSig").DownConversion.Enabled = True

.Signals("CapSig").DownConversion.Frequency = DownConvFreq ' frequency from

.Signals("CapSig").DownConversion.IntermediateFrequency = IFreq

' frequency to

.Signals("CapSig").DownConversion.AttenuatorEnabled = False

.Signals("CapSig").SampleRate = CapFs

```

.Sigals("CapSig").SampleSize = CapN
.Sigals("CapSig").VoltageRange = CapVRange * 10 ^ (9# / 20#) ' fixed
                        ' gain of 9dB with attenuator bypassed
.Sigals("CapSig").LoadSettings
End With

TheHdw.Wait 0.004

' Trigger the Capture
TheHdw.TurboACCApture(CapPin).Sigals("CapSig").Trigger
Dim CapDone As Boolean
' Wait for Capture to finish, stop Source, Disconnect Source and Capture
CapDone = TheHdw.TurboACCApture(CapPin).Sigals.WaitForCaptureDone(1)

' Release the StdCW from the Cap Down Converter and switch off attenuator
TheHdw.TurboACCApture(CapPin).DownConversion.Enabled = False
TheHdw.TurboACCApture(CapPin).DownConversion.AttenuatorEnabled = False

' Disconnect the Capture DUT Pins
TheHdw.TurboACCApture(CapPin).Disconnect tlTurboACCAptureFromPOS Or _
    tlTurboACCAptureFromNEG, tlTurboACCAptureToDUT Or _
    tlTurboACCAptureToCrossChannelLoopBack

' Bind capture to local reference
CapWave = TheHdw.TurboACCApture(CapPin).Sigals("CapSig").DSPWave

' Invoke DSP
CapBin = 821
rundsp.AnalyzeCapture CapWave, CapBin, Freq, Amp, Offset, SFDR, THD, _
    SNR, SINAD

' Test Limits
TheExec.Flow.TestLimit Freq, unit:=unitHz, forcereults:=tlForceFlow
TheExec.Flow.TestLimit Amp, unit:=unitVolt, forcereults:=tlForceFlow
TheExec.Flow.TestLimit Offset, unit:=unitVolt, forcereults:=tlForceFlow
TheExec.Flow.TestLimit SNR, unit:=unitDb, forcereults:=tlForceFlow
TheExec.Flow.TestLimit THD, unit:=unitDb, forcereults:=tlForceFlow
TheExec.Flow.TestLimit SINAD, unit:=unitDb, forcereults:=tlForceFlow

```

TheExec.Flow.TestLimit SFDR, unit:=unitDb, forcereults:=tlForceFlow
End Function

		
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[TurboAC Capture Examples](#) : [TurboAC Capture CW Down Conversion](#) : DSP Code

DSP Code

Public Function AnalyzeCapture(ByVal CapWave As DSPWave, ByRef CapBin As Double, _
ByRef Freq As Double, ByRef Amp As Double, _
ByRef Offset As Double, ByRef SFDR As Double, _
ByRef THD As Double, ByRef SNR As Double, _
ByRef SINAD As Double) As Long

Dim CapSpec As New DSPWave

Set CapSpec = CapWave.Spectrum

' Divide through by 9dB gain in downconverter when not using downconverter attenuator

' Set CapSpec = CapSpec.Divide(10 ^ (9# / 10))

Freq = CapSpec.CalcFrequencyFromSpectrum(CapBin)

Amp = CapSpec.CalcAmplitudeFromSpectrum(CapBin)

Offset = CapWave.CalcMean

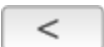
SFDR = CapSpec.CalcSFDR(CapBin)

THD = CapSpec.CalcTHD(CapBin, 5)

SNR = CapSpec.CalcSNR(CapBin, 5)

SINAD = CapSpec.CalcSINAD(CapBin)

End Function

	
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TurboACCaptureDefTopic

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About the TurboACCapture

About the TurboACCapture

The TurboACCapture instrument set provides AC waveform capture capabilities for testing devices such as precision A/D and D/A converters, high speed A/D and D/A converters and high performance SOC's such as Set Top Box devices.

Typical tests performed using the TurboAC instruments include gain accuracy, frequency response, distortion, and signal-to-noise ratio.

The TurboACCapture instrument can capture high-accuracy, low-noise and distortion signals with frequency components of up to 150MHz.

This section includes the following topics:

- | | |
|---|---|
| • | Features |
| • | Safety Information |
| • | Key Terms |
| • | TurboAC Capture Software Licensing Requirements |

TurboACCapture information includes the following sections:

- | | |
|---|--|
| • | TurboACCapture Theory of Operation |
| • | TurboACCapture Programming |

- [TurboACCapture Debug Display](#)

- [TurboAC Capture Examples](#)

Related Topics

- [TurboAC Specifications](#)

- [TurboACCapture](#) (VBT syntax)

- [TurboAC DIB Design Guidelines](#)

The TurboACCapture is a dual source per each TurboAC instrument board. Each board has two TurboACSource instruments as well. For more information, see [About the TurboACSource](#).

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About the TurboACCapture : Features

Features

The TurboACCapture instrument can accurately acquire differential signals with frequency components from DC to 15MHz. While the TurboACCapture’s internal processing is performed at 100Msps, the instrument’s DSP capabilities digitally resample the captured data to the user-specified sample rate in real time. The range of sample rates you can program is 5ksps to 50Msps. The TurboACCapture also has programmable voltage ranges and DC offset functionality for cancelling out common mode or DC bias voltages.

The TurboACCapture offers:

- | | |
|---|---|
| • | Selectable input ranges |
| • | Programmable sample rates of 5ksps to 50Msps |
| • | Programmable DC offset removal |
| • | Mixed-signal pattern microcode control |
| • | Access to Pin Parametric Measurement Unit (PPMU) per pin |
| • | Parameter sets (PSets) for pattern control of instrument settings |
| • | DIB Access connectivity |

For more information, see TurboAC Specifications.

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About the TurboACCapture : Safety Information

Safety Information

When operating a TurboACCapture instrument, always follow general UltraFLEX test system safety guidelines. For more information, see [Test System Safety](#).

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About the TurboACCapture : Key Terms

Key Terms

AUX input	The port used by either the Optional CW or Optional LO.
BBAC	Broadband Analog Channel
CCC	Channel Card Common
compatibility port	Port 1 of each TurboACSource and capture instrument instance. An instrument instance, whether source or capture, has four ports, the first of which is the compatibility port. The compatibility port can be programmed using BBAC syntax or TurboAC syntax. The other three ports of the instrument instance can only be programmed using TurboAC syntax.
CW	Continuous Waveform Source. CW is a capability that increases the TurboACSource bandwidth by allowing a synthesizer to be routed through the instrument delivery path.
DGS	Device ground sense
DIB LO	A user supplied signal that can be used as an alternative to the Standard or the Optional CW as the capture instrument’s local oscillator on the TurboAC board.
DIB Source	A user supplied source signal that can output to the TurboACSource ports.
direct capture mode	Allows capture and process of signals in a bandwidth from DC to 15MHz.

down conversion	A mixing technique used in a capture instrument to shift the tones being captured to another frequency. The basic capture instrument has a bandwidth of 15MHz. Used normally, a capture would contain tones between DC and 15MHz. Using the down conversion circuitry, a capture can be made of 10MHz sections of bandwidth of signals at higher frequencies (for example, the 10MHz band between 65 - 75MHz).
HSD	High Speed Digital
IIM	Instrument Initiated Move
ISL	Instrument Sync-Link
intermediate frequency	The frequency to which the target tone is shifted by a down conversion operation. If the target tone frequency is RF and the local oscillator frequency is LO, the intermediate frequency (IF) is $IF = LO - RF$. (This assumes high-side injection.)
ksps	Kilosamples per second
Local Oscillator	Generate a sinusoidal output used for down conversion.
Msp/s	Megasamples per second
MSW	Mixed Signal Workshop
Optional CW	An optional high frequency continuous waveform source that can be used with the TurboAC instrument. This frequency synthesizer can be of any brand. When connected to the TurboAC instrument, this frequency synthesizer would provide the optional CW signal for the source channel, and the optional LO signal for the capture channel. IG-XL provides programming and calibration support for the optional CW generator.
Optional LO	Alternate term for the Optional Frequency Synthesizer when used as a LO.
port	The set of pins that carries a signal to or from a location on the DIB. For the TurboACCapture, a port is composed of two pins: Positive and Negative. Each instrument instance has four capture ports labeled A, B, C, and D.
PPMU	Pin Parametric Measurement Unit
PSet	Parameter set

signal cancellation	A technique to suppress a tone in a captured signal. The signal cancellation technique requires a signal from the source instrument instance that is on the same board as the capture instrument instance. A signal cancellation application typically requires a pattern to control the relative phase between the canceling signal and the captured signal.
Standard CW	The default frequency synthesizer that is part of a TurboAC instrument. The PTS provides the continuous waveform signal for the capture channel's Internal LO local oscillator signal.
Turbo-Base channel card	This is the TurboACCapture back end circuitry, including the AWG and capture hardware.
Turbo-Frame channel card	The TurboACCapture analog front end circuitry, including: • Signal cancellation • Capture filter • Down convertor • Differential input mux

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About the TurboACCapture : TurboAC Capture Software Licensing Requirements

TurboAC Capture Software Licensing Requirements

TurboAC instrument hardware can support multiple frequencies across source and capture instruments. However, IG-XL enforces a software licensing scheme that constrains instrument functionality to a specified frequency limit across source and capture instruments. The TurboAC instrument can be configured with one of the following licenses:

Full License	A full license enables: - Full specification performance and functionality for two source and two capture instruments - AWG mode at 15MHz Optional CW is enabled by a separate license.
Single License	A single license enables: - TurboAC Frame card including 1 MX321 Turbo Channel card with Riders (one capture and one source instrument) 1 PTS310 synthesizer in the Support or Expansion cabinet - A Synthesizer Interface Module within the same u-space as the PTS

Audio License

- AWG mode at 15MHz

Optional CW source is enabled by a separate license.

An audio license enables:

- One capture and one source instrument
- AWG mode at 3MHz

CW source and down conversion features are not available with this license.

License Error Handling

IG-XL checks that the correct license and configuration are on the test system. If the license and configuration do not match, a run-time error is returned at validation time. The following table lists how IG-XL handles license and configuration mismatches:

License and Configuration Mismatch Handling			
Configuration	Full License	Single License	Audio License
Full Configuration	No Error.	Error: Turbo-CC-2 or PTS-2 is not licensed.	Error: PTS-1 or 2 is not licensed.
Single Configuration	Error: Trying to use Non-existing Turbo-CC-2 or PTS-2.	No Error.	Error: PTS-1 is not licensed. Error: Trying to use Non-existing Turbo-CC-2 or PTS-2.
Audio Configuration	Error: Trying to use Non-existing PTS-1 or 2.	Error: Turbo-CC-2 is not licensed. Error: Trying to use Non-existing PTS-1 (or 2).	No Error.

Related Topics

For more information, see IG-XL Licensing.

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TurboACCArchive Debug Display

TurboACCArchive Debug Display

The TurboACCArchive debug display is an interactive diagram of the TurboACCArchive hardware and signal path which shows the instrument’s settings and connections in the context of a site. You can change the values, options, and settings in the display and run your test program to see the effects of the changes.

The following sections describe the display’s controls and how to use the display:

- Starting the TurboACCArchive Debug Display
- Capture Signal
- Microcode Control
- Making Channel Connections

This section describes features specific to the TurboACCArchive debug display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

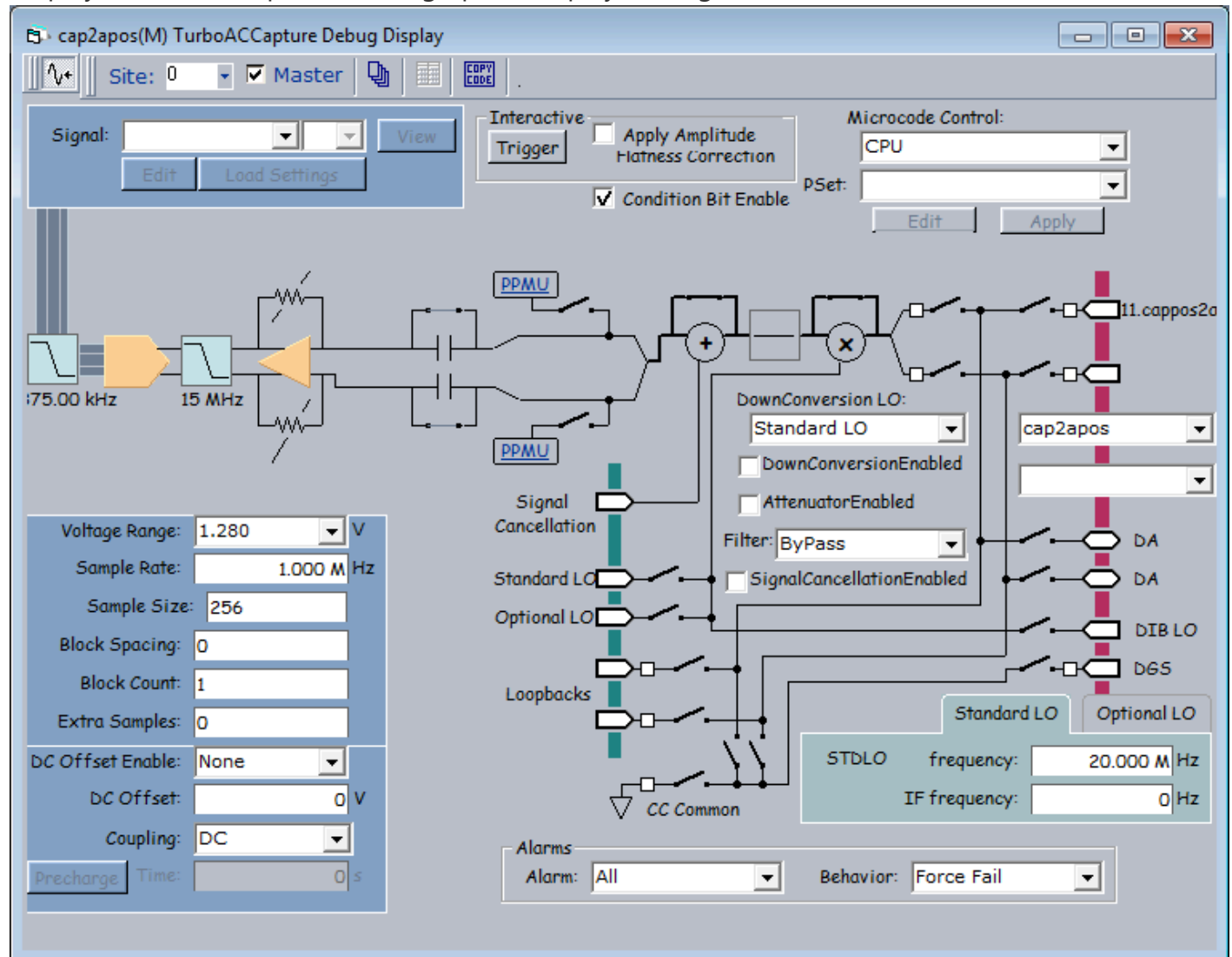
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TurboACCApture Debug Display : Starting the TurboACCApture Debug Display

Starting the TurboACCApture Debug Display

To launch the TurboACCApture Debug Display, first validate a test program that includes TurboACCApture pins. Then, on the IG-XL Tools tab in the Launch group, select Debug Displays>TurboACCApture to bring up the display (see figure below).



TurboACCArchive Debug Display

Incomplete Circuit or Errors Message when Starting the Debug Display

Does the TurboAC debug display connection matrix appear incomplete, or does one of the following error messages appear next to the toolbar?

! The test program has not been validated.

! No pin exists of type TurboACCArchive.

These problems indicate that the active test program was not validated before starting the debug display, or that the program does not include TurboACCArchive pins.

To correct these errors:

1.

Return to the IG-XL DataTool workbook and verify that TurboACCArchive pins are included and assigned in the pin map and channel map.
2.

 Click the Validate Job button.

If the test program successfully validates without errors, you can return to the TurboACCArchive debug display, which will now be complete and ready for use. If the test program does not validate successfully, you must correct all errors before the debug display can be complete.

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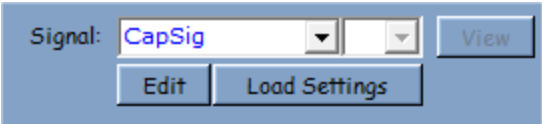
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TurboACCApture Debug Display : Capture Signal

Capture Signal

This block includes input boxes and buttons that select which signal to load and allows you to view the signal in WaveScope.



Capture Signal Control

The TurboACCApture signal parameters and buttons are:

Signal	All user-defined signal names appear in this field. When a signal is selected, the Load Settings button is enabled. When a capture is moved to the Support Board and it has not been redirected to a DSP Procedure, that signal's name is marked with an asterisk (*). You can view such a signal in WaveScope. If multiple captures on the Support Board are associated with the same Signal name, the box between the Signal field and the View button is enabled, and you can select the capture to view.
View	This button is enabled when you select a signal containing an asterisk. Viewing a signal only makes a copy of the samples on the Support Board. The data is not removed from the Support Board.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.

Load Settings

This button is enabled when a signal name is selected from the Signal field. It loads the settings of the specified Signal. That signal must be defined using VBT.

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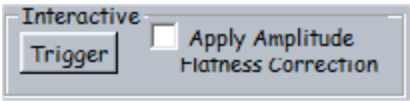
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TurboACCApture Debug Display : Capture Signal : Capture Trigger

Capture Trigger

This block controls triggering a capture and applying flatness correction:



Capture Trigger Control

The TurboACCApture trigger and flatness correction controls are:

Trigger	This button functions similarly to a single trigger button on an oscilloscope. You can set the sample rate, block count, block size, block spacing, and extra samples on the display and then click Trigger to make a capture. The trigger is disassociated from the Signal name. No signal must exist for the Trigger button to function. Captures made from the display are shown on WaveScope and the data is then discarded. The display does not move data to the Support Boards.
Apply Amplitude Flatness Correction	Applies extra processing on captured samples to improve the frequency based flatness of the capture. For the time domain $0 \propto 2 \cdot N \cdot \text{Log}_2 N + N$ For the frequency domain $0 \propto N \cdot \text{Log}_2 N + N$ With N being the captured sample size.

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TurboACCapture Debug Display : Microcode Control

Microcode Control

This section of the display includes controls to select the origin of the pattern microcode start command and a PSet to be applied later.

Microcode Control:

CPU

PSet:

Edit

Apply

Microcode Control

The parameters and buttons that control the microcode settings are:

Microcode Control	Specifies where the trigger command, which starts the capturing of the waveform, originates. The drop-down list options are: - CPU - External Trigger
PSet	Lists the parameter sets (PSets) that were defined in language. To apply a PSet from the debug display, you must have previously defined PSets using VBT. Otherwise, the drop-down list is empty.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor in the TDE section.
Apply	Allows you to apply a PSet from the PSet list.

Condition Bit Enable

When checked, the capture instrument generates a condition bit when the capture is complete.

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TurboACCApture Debug Display : Microcode Control : Capture Signal Setup

Capture Signal Setup

This area of the debug display controls the parameters of the capture signal.

Voltage Range:1.280V

Sample Rate:1.000 MHz

Sample Size:256

Block Spacing:0

Block Count:1

Extra Samples:0

DC Offset Enable:None

DC Offset:0V

Coupling:DC

Precharge Time:0s

Capture Signal Setup

The parameters that control signal setup are:

Voltage Range	The voltage range of the capture. You can enter any value between the minimum and maximum voltage range, but IG-XL sets the voltage range to the closest range above the programmed voltage.
Sample Rate	The number of samples to capture per second.
Block Size	The number of samples per block.
Block Spacing	The number of samples to skip per block. Minimum: 0

- [Maximum: 4 294 967 295](#)

Block Count	The number of blocks to capture.
Extra Samples	The number of extra samples per block.
DC Offset Enable	Assigns a DC offset to the Capture pins. The options are: • Both • Neg • None • Pos
DC Offset	The DC offset circuitry is added to the display according to the assignment selected.
Coupling	Sets the amount DC offset to apply.
Precharge Time	Selects either AC coupling or DC coupling mode.
PPMU	You can only precharge in AC coupling mode. The precharge and Precharge button are disabled in DC coupling mode. Click this button to open the PPMU debug display for the selected pin. See PPMU Debug Display.

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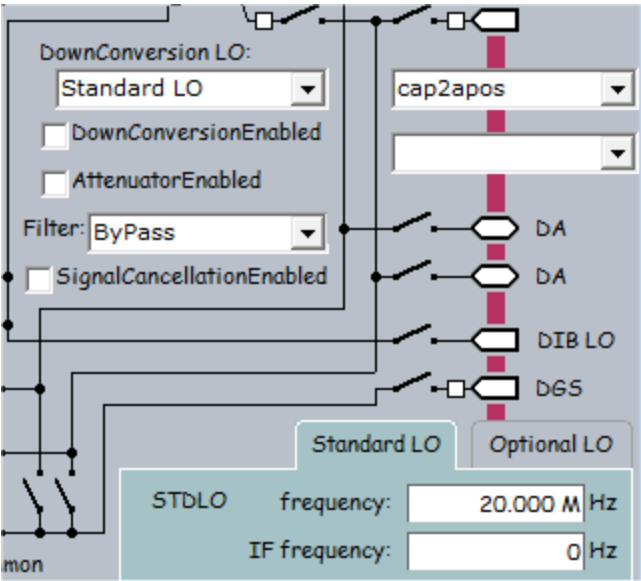
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TurboACCApture Debug Display : Microcode Control : Down Conversion and Signal Cancellation

Down Conversion and Signal Cancellation

This area of the debug display controls TurboACCApture down conversion features. You can set which oscillator is used and the parameters for the down conversion, as well as TurboACCApture filters and signal cancellation.



Down Conversion Control

The parameters that control down conversion are:

DownConversion LO	Use this property to get or set the enable state of the down conversion circuitry. When changing state from bypassed to enabled, the selected local oscillator source (DownConversion.LocalOscillatorSource) is switched in. If the selected local oscillator source is supported by IG-XL, it is programmed to the calculated local oscillator frequency (DownConversion.LocalOscillatorFrequency).
-------------------	--

	A run-time error is returned when enabling down conversion if the selected local oscillator is a shared resource which cannot be obtained (for example, it is already in use by the TurboACSource).
DownConversion Enabled	Enables the down conversion circuitry. Unchecking this control disables the down conversion circuitry.
AttenuatorEnabled	Enables the 9dB attenuator of the down conversion circuit. Unchecking this control disables the down conversion attenuator.
Filter	Sets the filters.
Signal Cancellation Enabled	Sets the enable state of the down conversion circuitry. When changing state from bypassed to enabled, the selected local oscillator source is switched in. If the selected local oscillator source is supported by IG-XL, it is programmed to the calculated local oscillator frequency. A run-time error is returned when enabling down conversion if the selected local oscillator is a shared resource which cannot be obtained (for example, it is already in use by the TurboACSource).
frequency	Gets or sets the down conversion frequency. This value is used with the .IntermediateFrequency to determine the .LocalOscillatorFrequency. The LocalOscillatorFrequency is set to the sum of the down conversion .Frequency property and the .IntermediateFrequency property. The down conversion .Frequency property always reads back the value you have programmed. If a supported local oscillator is selected by .LocalOscillatorSource, then the resulting .LocalOscillatorFrequency must lie within the source's capabilities. A run-time error is returned if the source cannot realize the required local oscillator frequency.
IF frequency	Gets or sets the intermediate frequency (IF) for TurboACCapture down conversion. This value is used with the DownConversion.Frequency to determine the local oscillator frequency. If a supported local oscillator is selected by .LocalOscillatorSource, then the resulting local oscillator frequency must lie within the source's capabilities. A run-time error is returned if the source cannot realize the required local oscillator frequency. Valid range for the .IntermediateFrequency property is -15MHz to 15MHz. Programming any value outside this range results in run-time error.

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TurboACCApture Debug Display : [Microcode Control](#) : Alarms

Alarms

Select an alarm and a behavior for that alarm using these controls:

Alarm

Although the drop-down list has two options, All and Overrange currently perform the same function since the TurboACCApture has only one alarm.

Behavior

Choose from the following menu:

- [Continue](#)
- [Default](#)
- [Force Bin](#)
- [Force Fail](#)
- [Off](#)
- [Raise Error](#)

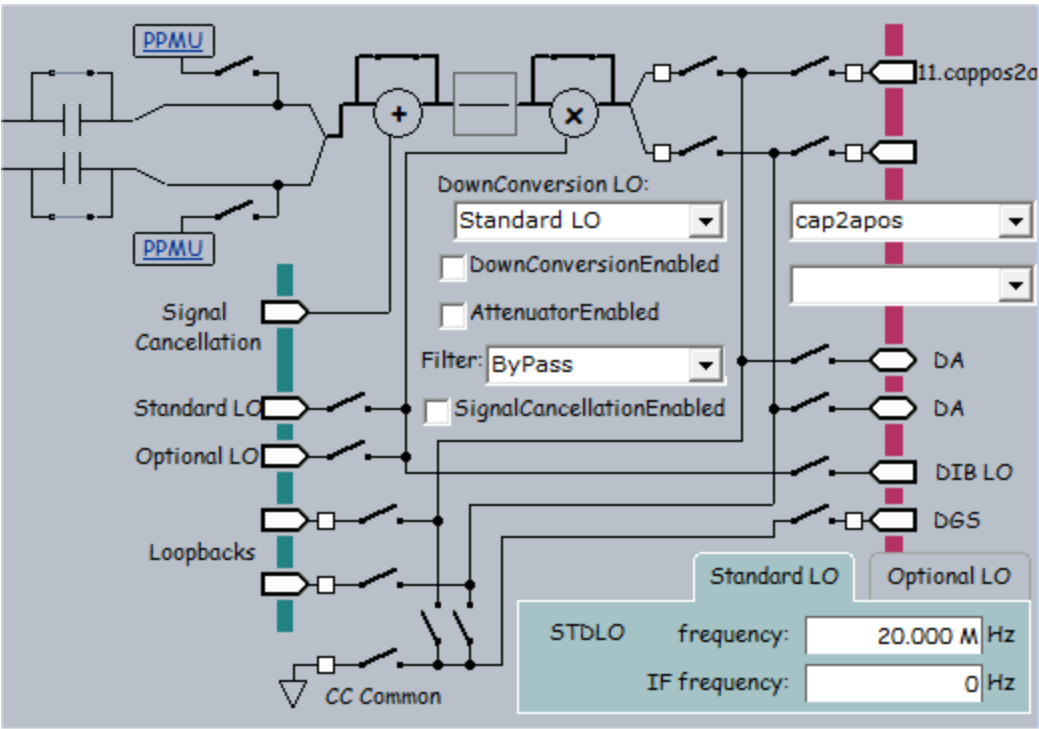
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TurboACCApture Debug Display : Making Channel Connections

Making Channel Connections

The figure shows the connections matrix on the TurboACCApture debug display. The matrix shows all possible TurboACCApture channel connections to available resources.



TurboACCApture Debug Display Connections Matrix

The following topics describe the interface behavior and the actions that you perform to make channel and resource connections in the TurboACCApture debug display:

- Opening and Closing a Relay Path
- Highlighting a Closed Relay Path

- [Determining a Local or Indirect Channel Connection](#)

The following sections describe the resources that the TurboACCAperture channels can be connected to:

- [Connecting DUT Pins](#)
- [Connecting DIB Access Channels](#)
- [Connecting to Device Ground Sense](#)
- [Connecting the Auxiliary Pin](#)
- [Connecting the Internal Analog Loopback Path](#)
- [Connecting a PPMU](#)
- [Connecting Channel Card Common](#)

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TurboACCApture Debug Display : Making Channel Connections : Opening and Closing a Relay Path

Opening and Closing a Relay Path

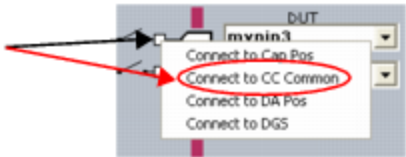
The TurboACCApture debug display simulates channel connections to the DUT and other resources. Graphically these connections are represented by open and closed relays along the connection path. However, clicking a relay does not open or close it. To open or close a relay path:

1.

Right-click the node along the signal path. A menu appears and lists the connect or disconnect options for that path.
2.

Select the connection destination from the menu. The relay opens or closes to make or break the connection.

Right-click the node and select the connect/disconnect option.



Connect/Disconnect Channels and Resources Menu

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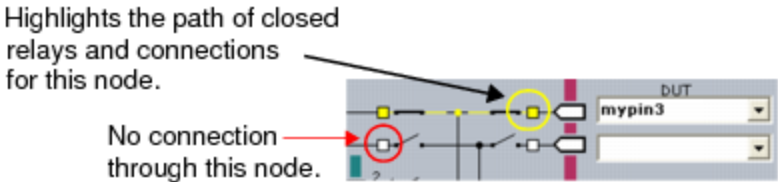
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TurboACCArchive Debug Display : Making Channel Connections : Highlighting a Closed Relay Path

Highlighting a Closed Relay Path

To help you determine which resources are connected to a specific TurboACCArchive channel, you can highlight a connection’s relay path.

The figure shows that you can highlight a closed relay path by left-clicking and holding the node along the closed relay path. The path of closed relays and connections that run through that node are then highlighted in yellow.



Tracing the Closed Relay Path

If no connection goes through that node, the node turns red.

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TurboACCApture Debug Display : Making Channel Connections : Determining a Local or Indirect Channel Connection

Determining a Local or Indirect Channel Connection

The debug display's connection matrix shows the connection paths related to TurboACCApture channels. These channel connections can be the result of connections made on this display or those made on another instrument's debug display.

For example, a TurboACCApture's DUT path may be connected to a PPMU. Although this type of connection can only be made from the PPMU display, it is reflected in both the TurboACCApture and PPMU debug displays.

When you click the DUT port of such a channel, a menu appears and lists all pins behind that channel. The symbol next to the pin name indicates how the connection was made. A check mark next to the pin name indicates that the connection is closed from the TurboACCApture instrument to the DUT. An asterisk is next to the pin name indicates that the connection was made as a consequence of another connection.

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TurboACCApture Debug Display : Making Channel Connections : Connecting DUT Pins

Connecting DUT Pins

The TurboACCApture debug display allows you to connect an instrument channel and other resources to the DUT pins specified in the test program’s channel map.

Starting the display automatically reads in the pins and lists them in the DUT pin drop-down lists.

The DIB channel string is also read into the display and is shown beside the instrument channel.

To connect an instrument channel or resource to a DUT pin:

- | | |
|----|---|
| 1. | Right-click the DUT node. |
| 2. | Select the destination resource or channel from the menu. |

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TurboACCApture Debug Display : Making Channel Connections : Connecting DIB Access Channels

Connecting DIB Access Channels

DIB access is a feature that allows a DUT pin to be connected to one instrument board through another instrument board. It is a method for connecting multiple resources on different boards to a single pin.

Since DIB access involves connecting between instruments, it also means coordinating connections between the two instrument’s debug displays. To distinguish one instrument from another, we refer to one as the primary instrument and the other as the secondary. The primary instrument is the one programmed to execute a test. The secondary instrument is the messenger or path that provides access to a specific pin.

In making a DIB access connection, you can only close the DIB Access connection from the primary instrument’s debug display.

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TurboACCApture Debug Display : Making Channel Connections : Connecting to Device Ground Sense

Connecting to Device Ground Sense

Device ground sense (DGS) is a signal that can be connected to the ground of the DUT. DGS is used as a lowside reference for the PPMUs.

To connect a DUT pin or instrument channel to the DGS:

- | | |
|----|--|
| 1. | Right-click the DGS node. |
| 2. | Select the destination pin or channel from the menu. |

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TurboACCapture Debug Display : Making Channel Connections : Connecting the Auxiliary Pin

Connecting the Auxiliary Pin

The auxiliary connection to the TurboACCapture allows another instrument to connect to the TurboAC. The auxiliary is not a general purpose instrument interconnection path. It is designed to work with cabling and future product options.

When the auxiliary connection is closed, the TurboACCapture is considered a secondary instrument and the primary instrument is using the TurboAC instrument. The auxiliary connections on the debug display are read-only and show whether the auxiliary connections are open or closed.

To open or close the auxiliary connection, make the connection from the primary instrument. Once the auxiliary connection is closed, no other connections on the TurboACCapture can be made.

When the auxiliary connection is closed, the Cap Pos pin is connected to AUX Pos and Cap Neg pin is connected to AUX Neg.

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TurboACCapture Debug Display : Making Channel Connections : Connecting the Internal Analog Loopback Path

Connecting the Internal Analog Loopback Path

The loopback connection is provided for debugging. It allows you to connect a TurboACSource directly to a TurboACCapture, allowing them to measure the sourced waveform.

The TurboAC instrument board has two instrument pairs (TurboACCapture and TurboACSource in Channel 1 and TurboACCapture and TurboACSource in Channel 2). The TurboACCapture Positive in Channel 1 loops back to the TurboACSource Positive of Channel 1. You cannot loop back instruments between the two channels (looping TurboACCapture Positive in Channel 1 with TurboACSource Positive of Channel 2).

You can control the Pos and Neg channels independently. For example, you can choose to loop back Positive, Negative, or Both.

To connect the TurboACSource to the TurboACCapture for loopback purposes:

1. Right-click the Loopback Positive node and select Connect to Cap Pos.
2. Right-click the Loopback Negative node and select Connect to Cap Neg.
3. Open the TurboACSource display.
4. Right-click the Loopback Positive node and select Connect to Src Pos.
5. Right-click the Loopback Negative node and select Connect to Src Neg.
6. Right-click the Source Reference node and select Connect to CCC.

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TurboACCApture Debug Display : Making Channel Connections : Connecting a PPMU

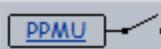
Connecting a PPMU

You can connect each TurboACCApture channel to a pin parametric measurement unit (PPMU) circuit for performing continuity tests.

Although the TurboACCApture debug display shows the channel connections to the PPMU circuits, you make PPMU connections in the PPMU debug display, not in the TurboACCApture display.

To connect a TurboACCApture channel (Pos, Neg) to a PPMU circuit:

1.



Click the PPMU hyperlink on the TurboACCApture debug display based on the TurboACCApture channel that is to be connected. The PPMU debug display starts.
2.

Select the pin to be connected to the PPMU from the Pin drop-down list.
3.

Click the relay to close the connection.

For more information, see PPMU Debug Display.

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TurboACCApture Debug Display : Making Channel Connections : Connecting Channel Card Common

Connecting Channel Card Common

You can connect a DUT pin to the channel card common (CCC) ground through either the positive or negative channels.

To connect a DUT pin or instrument channel to the CCC ground:

1. Right-click the CC Common node.
2. Select the destination pin or channel from the menu.

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TurboACCApture Programming

TurboACCApture Programming

This section describes programming of a TurboACCApture instrument to capture data from the DUT.

The following topics are available:

- | | |
|---|---|
| • | Simulated Configuration Files |
| • | TurboACCApture DataTool Programming |
| • | TurboACCApture Connections |
| • | Programming Down Conversion |
| • | Programming Input Coupling |
| • | Programming Voltage Ranging |
| • | Starting a TurboACCApture |
| • | Visual Basic for Test |
| • | TurboACCApture Reset Values |

- [TurboACCapture Pattern Control](#)

- [Using TurboAC in Concurrent Testing](#)

For more information, see:

- [TurboACCapture](#) (VBT Syntax Reference)

- [TurboAC Capture Examples](#)

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TurboACCapture Programming : Simulated Configuration Files

Simulated Configuration Files

You can use TurboAC Capture offline by forcing IG-XL to map the TurboAC instrument. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the TurboAC instrument in slots 11 and 23:
UltraFLEX IG-XL hardware configuration
This file contains an example hardware configuration, and
is used as the default offline.

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot] Type      idprom (type serial rev company)

-1.0      dib      239-001-00 0000000 0000-A 0000

1.0      HSD-800    979-219-00 0000000 0000-A 0000
```

	HSD-800	956-361-00 0000000 0000-A 0000
6.0	DC-30	805-002-00 0000000 0000-A 0000
	DC-30	949-901-00 0000000 0000-A 0000
3.0	GigaDigHD	979-999-00 0000000 0000-A 0000
	GigaDigHD	956-436-00 0000000 0000-A 0000
11.0	Turbo-Frame	974-213-01 03017D1A 0549-B 5445
	aka: 974-213-00	
	Turbo-Frame	939-367-01 03014072 0541-B 5445
	Turbo-CC-1	939-321-00 030172BE 0544-C 5445
	PowerBoard	939-364-00 03019B50 0528-B 5445
	Turbo-CC-2	939-321-00 030172C8 0544-C 5445
11.1	Turbo-SIM	974-228-01 03014213 0546-B 5445
	aka: 974-228-00	
	Turbo-SIM	939-366-01 0301570F 0542-B 5445
	aka: 939-366-00	
12.0	BBAC-15	979-214-00 0000000 0000-A 0000
	BBAC-15	956-375-00 0000000 0000-A 0000
23.0	Turbo-Base	974-215-00 03013B33 0549-D 5445
	Turbo-Base	956-375-14 03013A23 0549-D 5445
	aka: 956-078-00	
	Turbo-BaseRider	949-987-02 0301668A 0451-D 5445
	PowerBoard	949-984-04 03010C72 0502-D 5445
	aka: 949-984-00	
24.0	SupportBoard	974-220-12 0000000 0000-A 0000
	SupportBoard	956-354-00 0000000 0000-A 0000

</TEST_HEAD TH 1> t

To use the TurboAC instrument offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the TurboAC instrument online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see [Tester Configuration](#).

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[TurboACCApture Programming](#) : TurboACCApture DataTool Programming

[TurboACCApture DataTool Programming](#)

This section describes how to program several DataTool worksheets to handle basic TurboACCApture parameters. It includes the following topics:

- [Programming the Pin Map](#)
- [Programming the Channel Map](#)
- [Creating a Timing Set Using the Mixed Signal Workshop](#)

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TurboACCapture Programming : TurboACCapture DataTool Programming : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. You can enter any pin type for a TurboAC instrument. The recommended selection for TurboAC instruments is Analog.

For more information, see [Pin Map Sheet](#) in the DataTool section.

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TurboACCCapture Programming : TurboACCCapture DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the TurboACCCapture instrument, you must select one of four channel types from the Type column menu for each pin or pin group. Valid TurboACCCapture channel types are:

BBACCapNeg	BBAC Capture Negative Pin
BBACCapPos	BBAC Capture Positive Pin
TurboACCCapNeg	TurboACCCapture Negative Pin.
TurboACCCapPos	TurboACCCapture Positive Pin.

See [Specifying Tester Channels](#) in the DataTool section on how to determine the tester channel information value. The Signal naming conventions are:

- <slot #>.cappos<channel 1 a - d> (positive pin, for example: 11.cappos1a)
 - <slot #>.cappos<channel 2 a - d> (positive pin, for example 11.cappos2c)
 - <slot #>.capneg<channel 1 a - d> (negative pin, for example 11.capneg1b)
 - <slot #>.capneg<channel 2 a - d> (negative pin, for example 11.capneg2d)
- For more information, see [Channel Map Sheet](#) in the DataTool section.

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TurboACCAp Program : TurboACCAp DataTool Programming : Creating a Timing Set Using the Mixed Signal Workshop

Creating a Timing Set Using the Mixed Signal Workshop

The Mixed Signal Workshop (MSW) tool helps you to build reusable mixed signal timing solutions that meet coherence requirements.

The output of the MSW is a mixed signal timing set which contains key capture sampling parameters such as sample rate and sample size. Timing sets are stored on a Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests. For more information, see Mixed Signal Workshop.

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[TurboACCApture Programming](#) : TurboACCApture Connections

TurboACCApture Connections

The following topics describe the options for connecting a TurboACCApture instrument to various resources through the DIB:

- [Connecting to the DUT Through TurboACCApture Port](#)
- [Connecting to the Channel Map Specified DIB Site](#)
- [Connecting to a Selected Port](#)
- [Connecting Individual Pins to a Port](#)
- [BBAC Compatible Connections](#)
- [Making a Differential Connection to Capture Cancellation Circuitry](#)
- [Making a Single-Ended Connection to Capture Cancellation Circuitry](#)
- [Connecting to CC Common](#)
- [Connecting a Pin to DGS](#)
- [Connecting the Loopback Circuitry](#)

- [Connecting the Cross Channel Loopback Circuitry](#)
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- [Connecting to a PPMU](#)
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- [Connecting to DIB Access](#)

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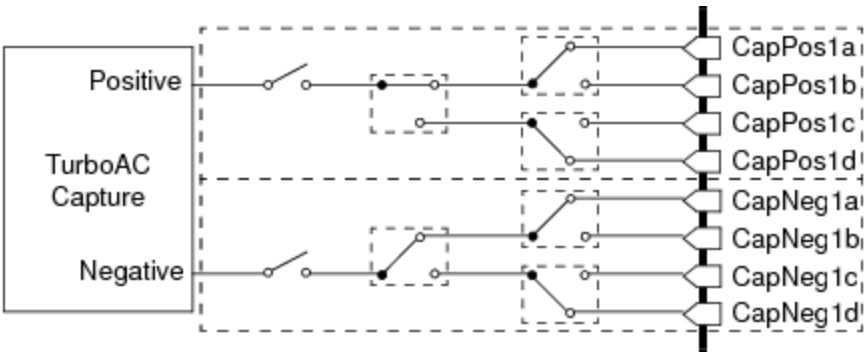
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TurboACCapture Programming : TurboACCapture Connections : Connecting to the DUT Through TurboACCapture Port

Connecting to the DUT Through TurboACCapture Port

Each TurboACCapture has four ports per the positive and negative pins as shown in the figure:



TurboACCapture Multiplexed Channels

A TurboACCapture can connect to any of the four Ports, but only one of the Ports at the same time. If any pins are connected to a Port, then none of the other Ports can be used at that time.

Entries in the channel map and connections in VBT can differ depending on what connections to the DUT are required.

For connections made using the debug displays, see Using the TurboACSource Debug Display.

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TurboACCApture Programming : TurboACCApture Connections : Connecting to the Channel Map Specified DIB Site

Connecting to the Channel Map Specified DIB Site
You can connect both the positive and negative pins using a single channel map entry. The following channel map example shows both the positive and negative pin for channel 1 PortA, but the following VBT syntax uses only the positive pin to connect both pins.

Channel Map for Basic Connections				
Pin Name	Package	Type	Site	Comment
PosPin		TurboACCApPos	18.cappos1a	
NegPin		TurboACCApNeg	18.capneg1a	

This VBT syntax connects both positive and negative pins to the DUT through PortA as specified in the Site column entry:
TheHdw.TurboACCApture.Pins("PosPin").Connect tlTurboACCAptureFromInst _
 , tlTurboACCAptureToDUT
tlTurboACCAptureFromInst determines that all pins for the channel are connected.
tlTurboACCAptureToDUT determines that the Port specified in the channel map is connected.

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TurboACCapture Programming : TurboACCapture Connections : Connecting to a Selected Port

Connecting to a Selected Port

You can connect a TurboACCapture to a port other than the one specified in the channel map Site column. Do this using a channel map and syntax similar to the following:

Channel Map for Basic Connections

Pin Name	Package	Type	Site	Comment
PosPin		TurboACCapPos	18.cappos1a	
NegPin		TurboACCapNeg	18capneg1a	

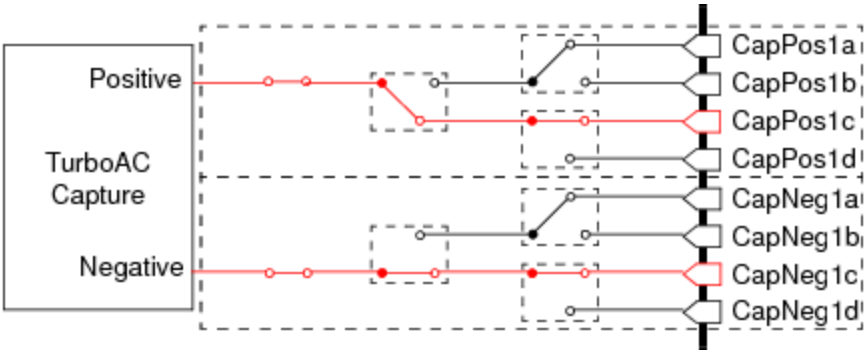
'Connect both pins to the DUT on Port C

```
TheHdw.TurboACCapture.Pins("PosPin").Connect tlTurboACCaptureFromInst _  
    , tlTurboACSourceToPortC
```

tlTurboACCaptureFromInst determines that all pins for the channel are connected.

tlTurboACCaptureToPortC determines that the pins are connected to port C, although port A is specified in the in the channel map.

The following figure shows what connections would be made to the DUT.



Connecting to a Specified Port

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TurboACCApture Programming : TurboACCApture Connections : Connecting Individual Pins to a Port

Connecting Individual Pins to a Port

For certain tests, you may not want to connect both TurboACCApture pins to the DUT at the same time. To do this each pin requires a separate entry in the channel map as shown in the following example:

Channel Map for Basic Connections

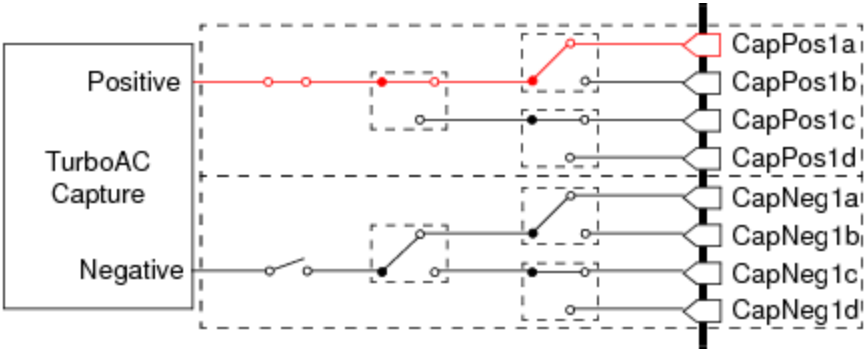
Pin Name	Package	Type	Site	Comment
PosPin		TurboACCApPos	18.cappos1a	
NegPin		TurboACCApNeg	18.capneg1a	

The following VBT statement connects the positive and negative pin to the DUT but leaves the reference disconnected:

' Connect the positive pin to Port A

```
TheHdw.TurboACCApture.Pins("PosPin").Connect tITurboACCAptureFromPOS _  
    , tITurboACCAptureToPortA
```

The figure shows what connections are made to the DUT.



Port Connections with Reference Disconnected

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TurboACCApture Programming : TurboACCApture Connections : BBAC Compatible Connections

BBAC Compatible Connections

You can use the TurboACCApture in a BBACCapture compatible mode. When programmed in this mode, only the A port can be used and all other ports are locked. The following shows a channel map configured for BBAC compatible mode:

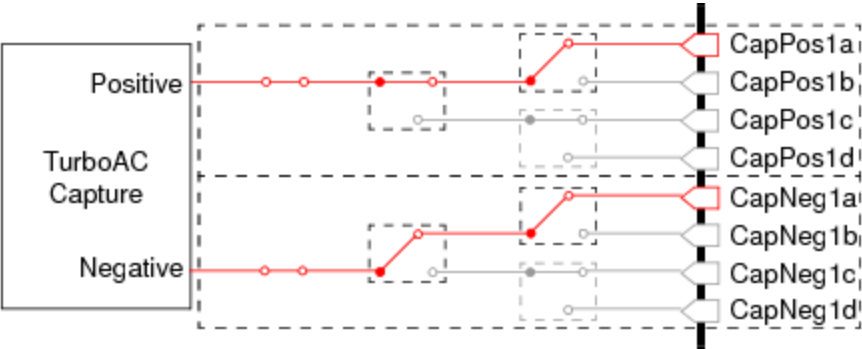
Channel Map Example of BBAC Compatible Mode

Pin Name	Package	Type	Site	Comment
PosPin		BBACCapPos	18.cappos1	
NegPin		BBACCapNeg	18.capneg1	

When using the TurboACCApture in BBAC compatible mode, you program the instrument as if it were a BBAC instrument. The Pogo site location used is that of the Turbo Frame board, but the BBAC type, VBT syntax, and debug displays are used to program the instrument. The following statement connects the TurboACCApture in BBAC compatible mode for the above channel map:

```
TheHdw.BBACCapture.Pins("PosPin").Connect t1BBACCaptureFromInst _  
    , t1BBACCaptureToDUT
```

The following figure shows how the ports are used in BBAC compatible mode.



TurboACCapture Connections in BBAC Compatible Mode

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TurboACCApture Programming : TurboACCApture Connections : Making a Differential Connection to Capture Cancellation Circuitry

Making a Differential Connection to Capture Cancellation Circuitry

You can connect the TurboACSource to the cancellation circuitry of the TurboACCApture instrument. See [Signal Cancellation](#) for a description of this feature.

During a differential signal cancellation, the TurboACSource connects to the TurboACCApture cancellation circuitry and the TurboACCApture connects differentially to the DUT.

Channel Map for Capture Cancellation

Pin Name	Package	Type	Site	Comment
CapPosPin		TurboACCApPos	18.cappos1a	
CapNegPin		TurboACCApNeg	18.capneg1a	
SrcPosPin		TurboACSrcPos	18.srcpos1a	
SrcNegPin		TurboACSrcNeg	18.srcneg1a	

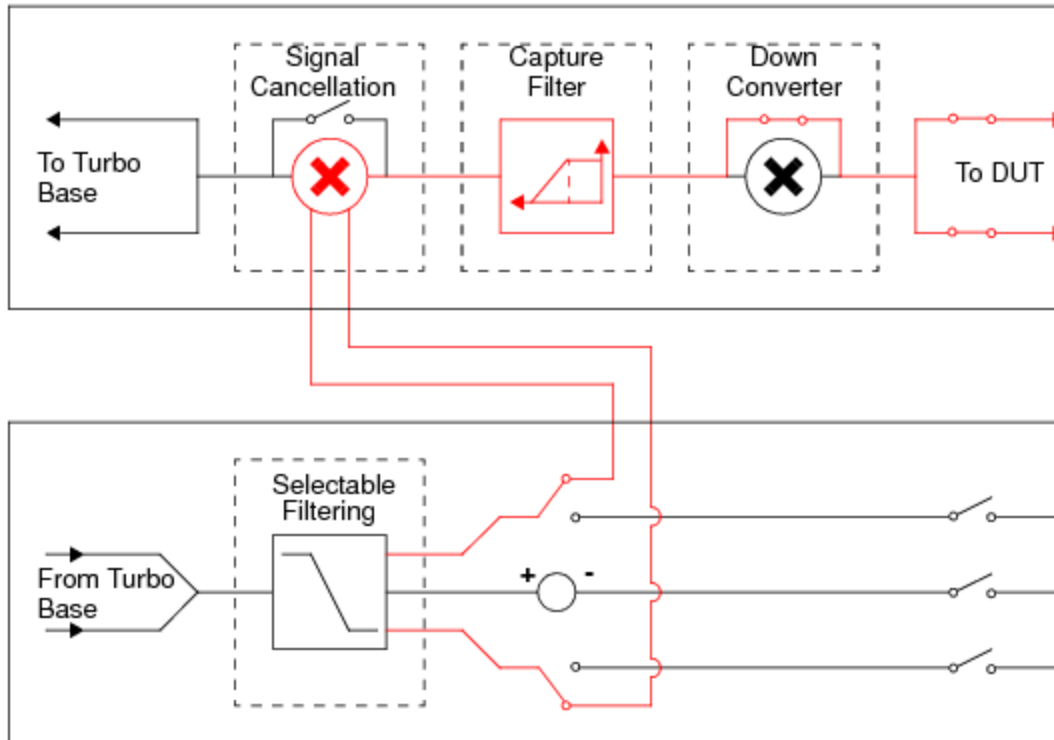
The following VBT statements make the required connections:

```
' Connect the TurboACSource to the TurboACCApture
TheHdw.TurboACSource.Pins("SrcPosPin").Connect tlTurboACSourceFromInst _
    , tlTurboACSourceToCaptureCancellation
' Enable the TurboACCApture cancellation circuitry
TheHdw.TurboACCApture.Pins("CapPosPin").SignalCancellationEnabled = True
' Connect the TurboACSource to the DUT allowing the signal to be input
TheHdw.TurboACCApture.Pins("CapPosPin").Connect tlTurboACCAptureFromInst,
tlTurboACCAptureToDUT
```

✓ Note:

The TurboACSource must connect differentially to the cancellation circuitry. You cannot connect only the positive pin or the negative pin. However, you can cancel a single-ended signal from the DUT. See Making a Single-Ended Connection to Capture Cancellation Circuitry.

The figure shows the connections between the TurboACSource and TurboACCapture in a differential signal cancellation setup. Note that the TurboACCapture has both the positive and negative pin connected.



TurboACSource Differential Connection to Signal Cancellation

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TurboACCArchive Programming : TurboACCArchive Connections : Making a Single-Ended Connection to Capture Cancellation Circuitry

Making a Single-Ended Connection to Capture Cancellation Circuitry

You can connect the TurboACSource to the cancellation circuitry of the TurboACCArchive instrument. See [Signal Cancellation](#) for a description of this feature.

During a single-ended signal cancellation, the TurboACSource connects to the TurboACCArchive cancellation circuitry differentially, and the TurboACCArchive connects the required pin to the DUT and the other pin to the CC Common.

Channel Map, Single-Ended Capture Cancellation				
Pin Name	Package	Type	Site	Comment
CapPosPin		TurboACCArCapPos	18.cappos1a	
CapNegPin		TurboACCArCapNeg	18.capneg1a	
SrcPosPin		TurboACSrcPos	18.srcpos1a	
SrcNegPin		TurboACSrcNeg	18.srcneg1a	

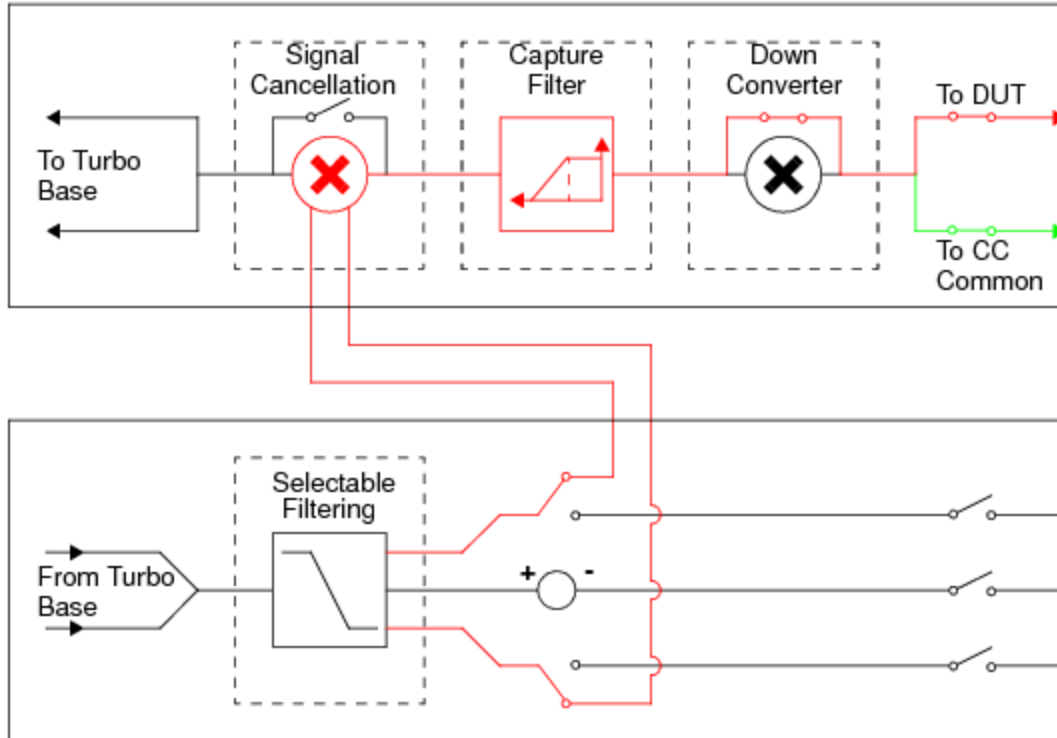
The following VBT statements make the required connections:

```
' Connect the TurboACSource to the TurboACCArchive
TheHdw.TurboACSource.Pins("SrcPosPin").Connect tlTurboACSourceFromInst _
    , tlTurboACSourceToCaptureCancellation
' Enable the TurboACCArchive cancellation circuitry
TheHdw.TurboACCArchive.Pins("CapPosPin").SignalCancellationEnabled = True
' Connect the TurboACSource to the DUT allowing the signal to be input
```

TheHdw.TurboACCapture.Pins("CapPosPin").Connect tITurboACCaptureFromPOS,
tITurboACCaptureToCCC

Note:

The TurboACSource must connect differentially to the cancellation circuitry. You cannot connect only the positive pin or the negative pin. However, you can cancel a single-ended signal from the DUT.



TurboACSource Connected to Signal Cancellation

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TurboACCApture Programming : TurboACCApture Connections : Connecting to CC Common

Connecting to CC Common

You can connect the TurboACCApture to the CC Common (Channel Card Common). Connections to CC Common can be made from positive and negative pins from any of the four ports. Only a single pin from a channel can be connected to CC Common at one time, and the pin connected is based on the DIB Site Location in the channel map. The following describes the connection types to CC Common:

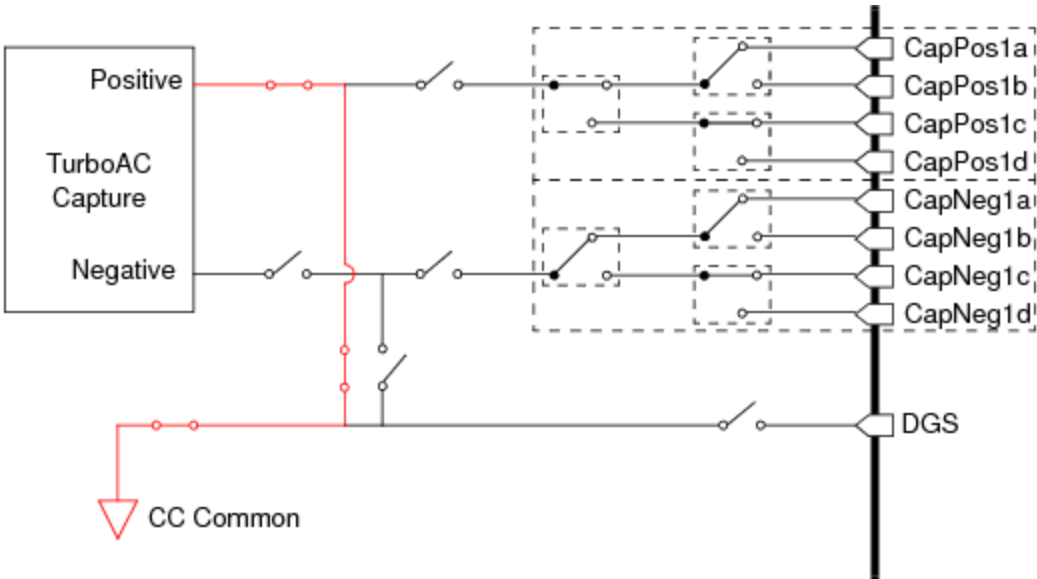
- Connecting a Capture Pin to CC Common
- Connecting a DUT Pin to CC Common

Connecting a Capture Pin to CC Common

You can connect a positive or negative pin from a TurboACCApture to the CC Common. Do this using a channel map and syntax similar to the following:

Channel Map for Connecting Capture Pin to CC Common				
Pin Name	Package	Type	Site	Comment
MyPin		TurboACCApPos	18.cappos1a	

TheHdw.TurboACCApture.Pins("MyPin").Connect tITurboACCAptureFromPOS _
, tITurboACSourceToCCC
tITurboACCAptureFromPOS determines that the capture positive pin for the channel is connected.
tITurboACCAptureToCCC determines that the pin is connected to CCC.
The following figure shows what connections would be made:



Connection to CC Common

Connecting a DUT Pin to CC Common

You can connect a pin through a TurboACCapture pin to the CC Common. Do this using a channel map and syntax similar to the following:

Channel Map for Connecting DUT Pin to CC Common

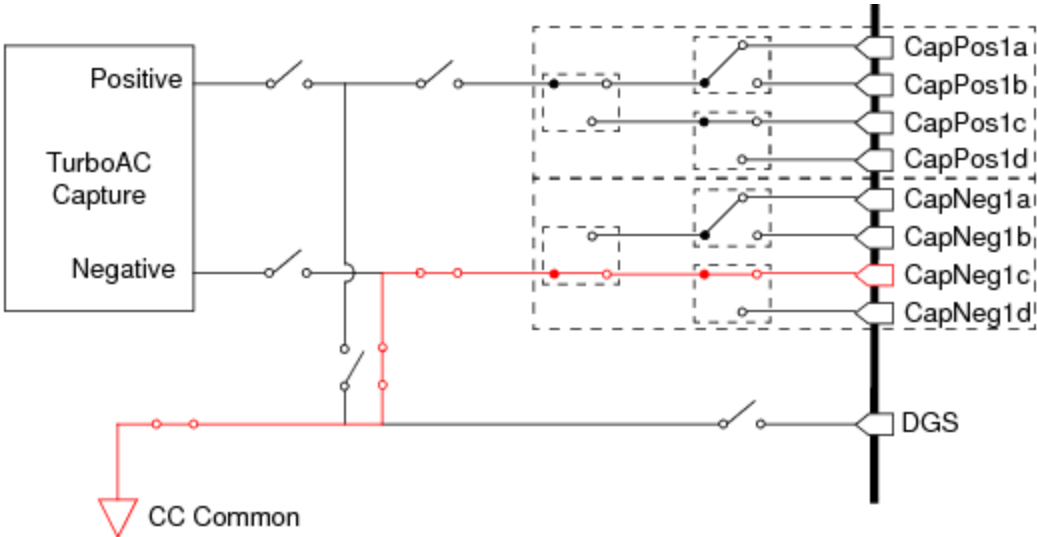
Pin Name	Package	Type	Site	Comment
MyPin		TurboACCapNeg	18.capneg1c	

TheHdw.RefGround.Pins("MyPin").Connect tlCCGroundTypeCCC

tlCCGroundTypeCCC determines that capture negative port 1C is connected for the channel.

The port connected is determined by the channel map DIB Site location site.

The following figure shows what connections would be made.



Connection to CC Common

The TurboACCapture does not support the following syntax:

TheHdw.RefGround.Pins(SrcPin).Connect tlCCGroundTypeDGS

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TurboACCapture Programming : TurboACCapture Connections : Connecting a Pin to DGS

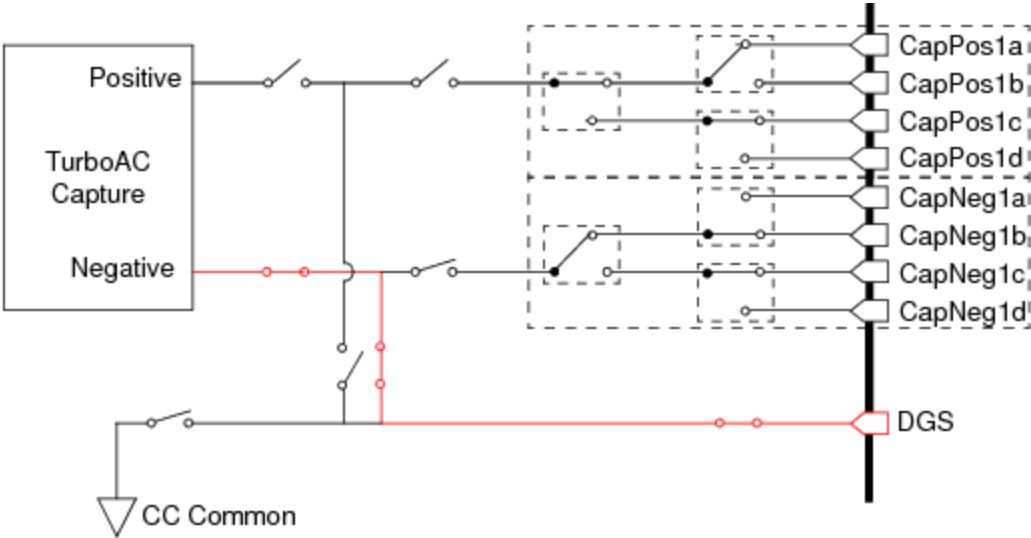
Connecting a Pin to DGS

You can connect a TurboACCapture pin to the DGS. Do this using a channel map and syntax similar to the following:

Channel Map for Connecting a Pin to DGS

Pin Name	Package	Type	Site	Comment
MyPin		TurboACCapNeg	18.capneg1b	

TheHdw.TurboACCapture.Pins("MyPin").Connect tITurboACCaptureFromNEG _
 , tITurboACCaptureToDGS
tITurboACCaptureFromNEG determines that the capture negative pin for the channel is connected.
tITurboACCaptureToDGS determines that the pin is connected to DGS.
The following figure shows what connections would be made.



DUT Pin Connected to DGS

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TurboACCapture Programming : TurboACCapture Connections : Connecting the Loopback Circuitry

Connecting the Loopback Circuitry

The TurboAC instrument supports internal analog loopback between the source and capture. For more information, see Internal Loopback Circuitry.

A standard loopback connection is made between source1 and capture1 or source2 and capture2. The loopback connection is made using the channel map specified DIB location. The Port that is specified does not matter for this connection, although the channel number specified does.

Channel Map for Basic Connections

Pin Name	Package	Type	Site	Comment
MySrcPin		TurboACSrcNeg	18.srcneg1c	
MyCapPin		TurboACCapNeg	18.capneg1c	

This VBT syntax connects source1 and capture1 in loopback:

' Connect both the positive and negative capture pins

```
TheHdw.TurboACCapture.Pins("MyCapPin").Connect tlTurboACCaptureFromInst, _  
tlTurboACCaptureToLoopBack
```

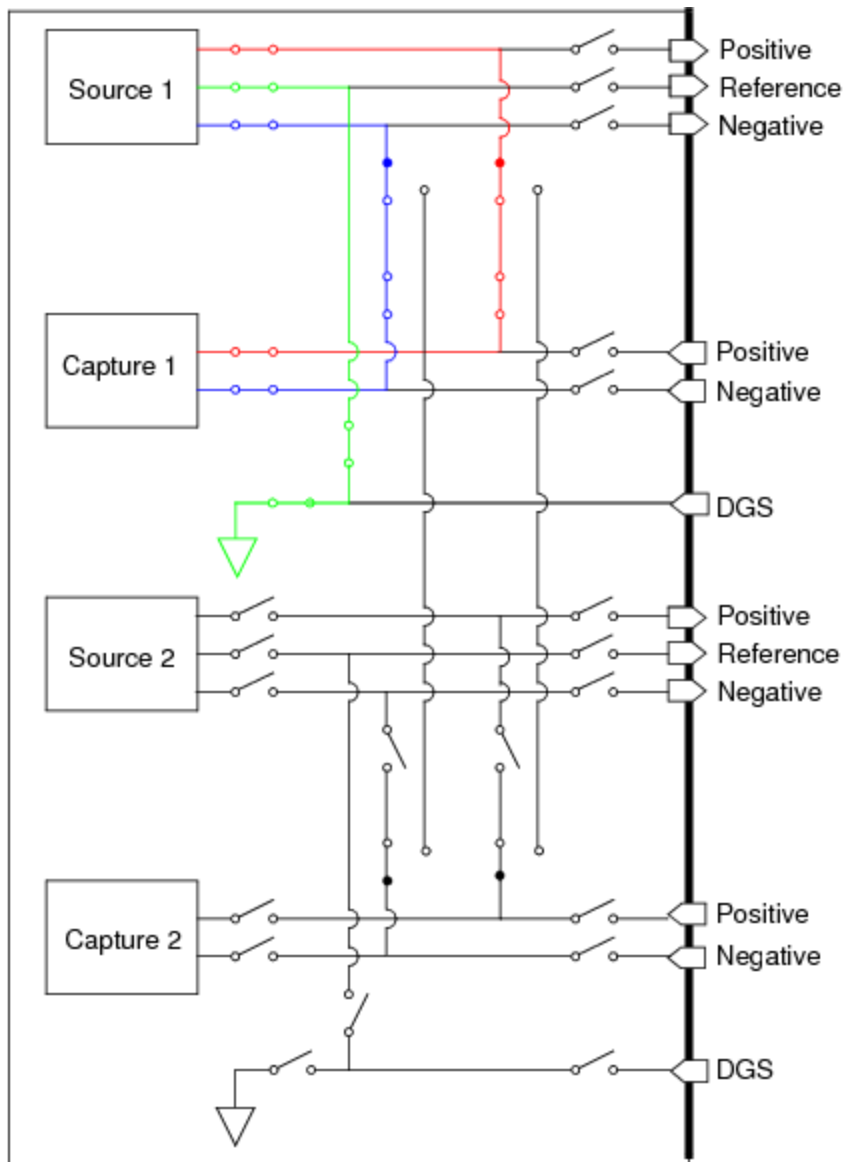
' Connect the positive and negative source pins, reference is not connected

```
TheHdw.TurboACSource.Pins("MySrcPin").Connect tlTurboACSourceFromNEG Or _  
tlTurboACSourceFromPOS, tlTurboACSourceToLoopBack
```

' Connect reference to CC Common

```
TheHdw.TurboACSource.Pins("MySrcPin").Connect tlTurboACSourceFromREF, _  
tlTurboACSourceToCCC
```

The figure shows what connections would be made.



Standard Loopback Connection

The following code shows disconnecting the TurboACSource and TurboACCapture from loopback:

' Disconnect both the positive and negative capture pins

```
TheHdw.TurboACCapture("MyCapPin").Disconnect tITurboACCaptureFromInst, _
    tITurboACCaptureToLoopBack
```

' Disconnect the positive and negative source pins

```
TheHdw.TurboACSource("MySrcPin").Disconnect tITurboACSourceFromNEG Or _
    tITurboACSourceFromPOS, tITurboACSourceToLoopBack
```

' Disconnect reference from CC Common

```
TheHdw.TurboACSource("MySrcPin").Disconnect tITurboACSourceFromREF, _
    tITurboACSourceToCCC
```

☑ Note:

Ensure that the TurboACSource and Capture are disconnected from other signal paths (for example, from the DIB/DUT) to avoid external signal interference while using internal loopback.

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TurboACCapture Programming : TurboACCapture Connections : Connecting the Cross Channel Loopback Circuitry

Connecting the Cross Channel Loopback Circuitry

The TurboAC instrument supports internal cross channel analog loopback between the source and capture. For more information, see Internal Cross Channel Loopback.

A cross channel loopback connection is made between source1 and capture2 or source2 and capture1.

The loopback connection is made using the channel map specified DIB location. The Port that is specified does not matter for this connection, although the channel number specified does.

Channel Map for Basic Connections

Pin Name	Package	Type	Site	Comment
MySrcPin		TurboACSrcNeg	18.srcneg1c	
MyCapPin		TurboACCapNeg	18.capneg2a	

This VBT syntax connects source1 and capture2 in cross channel loopback:

' Connect both the positive and negative capture pins

```
TheHdw.TurboACCapture.Pins("MyCapPin").Connect tlTurboACCaptureFromInst, _  
tlTurboACCaptureToCrossChannelLoopBack
```

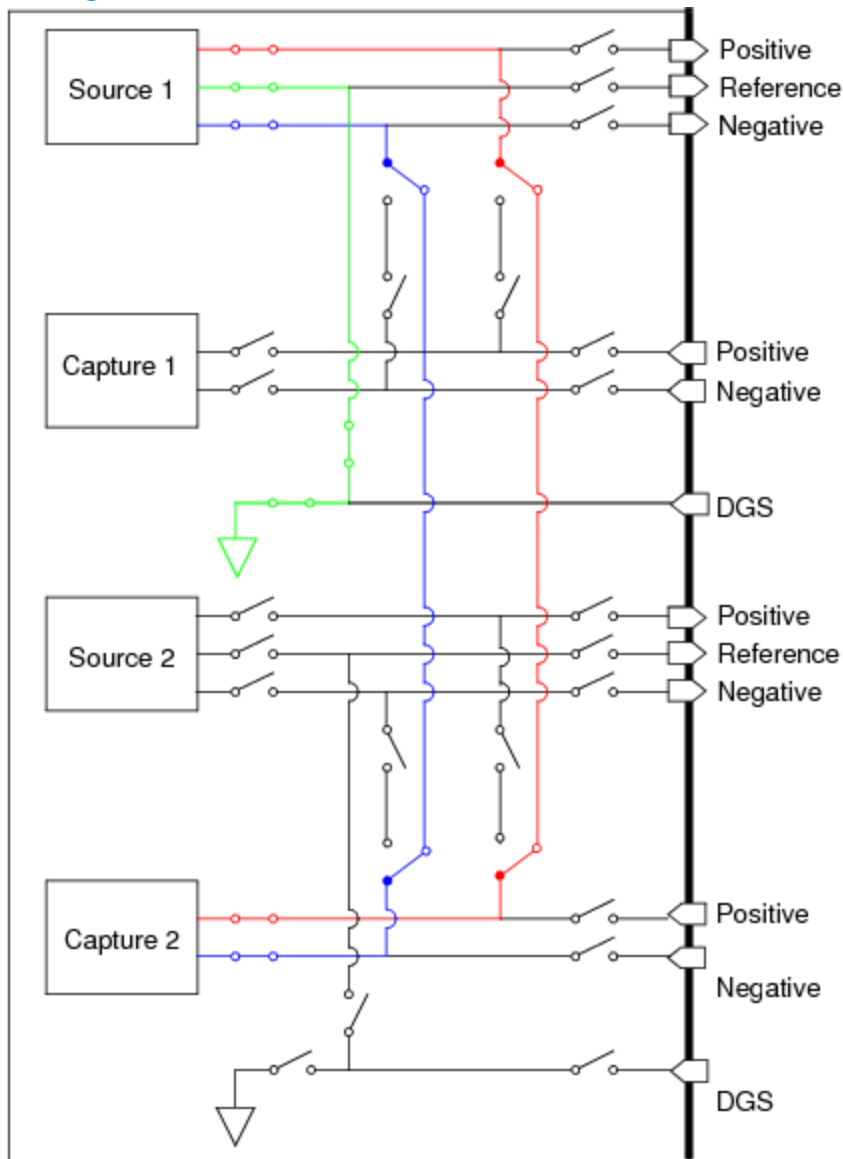
' Connect the positive and negative source pins, reference is not connected

```
TheHdw.TurboACSource.Pins("MySrcPin").Connect tlTurboACSourceFromNEG Or _  
tlTurboACSourceFromPOS, tlTurboACCaptureToCrossChannelLoopBack
```

' Connect reference to CC Common

```
TheHdw.TurboACSource.Pins("MySrcPin").Connect tlTurboACSourceFromREF, _  
tlTurboACSourceToCCC
```

The figure shows what connections would be made.



Cross Channel Loopback Connection

The following code shows disconnecting the TurboACSource and TurboACCapture from loopback:

' Disconnect both the positive and negative capture pins

```
TheHdw.TurboACCapture("MyCapPin").Disconnect tITurboACCaptureFromInst, _
    tITurboACCaptureToCrossChannelLoopBack
```

' Disconnect the positive and negative source pins

```
TheHdw.TurboACSource("MySrcPin").Disconnect tITurboACSourceFromNEG Or _
    tITurboACSourceFromPOS, tITurboACCaptureToCrossChannelLoopBack
```

' Disconnects reference from CC Common

```
TheHdw.TurboACSource("MySrcPin").Disconnect tITurboACSourceFromREF, _
    tITurboACSourceToCCC
```

✓ **Note:**

Ensure that the TurboACSource and Capture are disconnected from other signal paths (for example, from the DIB/DUT) to avoid external signal interference while using internal loopback.

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TurboACCApture Programming : TurboACCApture Connections : Connecting to a PPMU

Connecting to a PPMU

You can connect each TurboACCApture channel to a Pin Parametric Measurement Unit (PPMU) circuit for performing continuity tests.

Make connections when the TurboACCApture is in ByPass mode using the following syntax:

TheHdw.PPMU.Pins(PinList).Connect

The PinList can be a pin group of TurboACCApture pins, PPMU pins, or a combination of both.

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TurboACCApture Programming : TurboACCApture Connections : Connecting to DIB Access

Connecting to DIB Access

A test often requires several instrument types to be connected to a DUT pin in a test program. IG-XL supports this requirement in two ways:

- Directly connecting multiple resources to the DUT pin
- Using the DIB access feature to connect the DUT pin to one instrument board through another instrument board

When a device pin needs multiple or different resources, you can use DIB access to minimize the number of DIB relays that are required and to facilitate a simpler programming model. Once the channel map is set up to indicate the DIB connectivity, IG-XL addresses the correct resource based solely on the pin name. You need not create a virtual pin to reference the different resources.

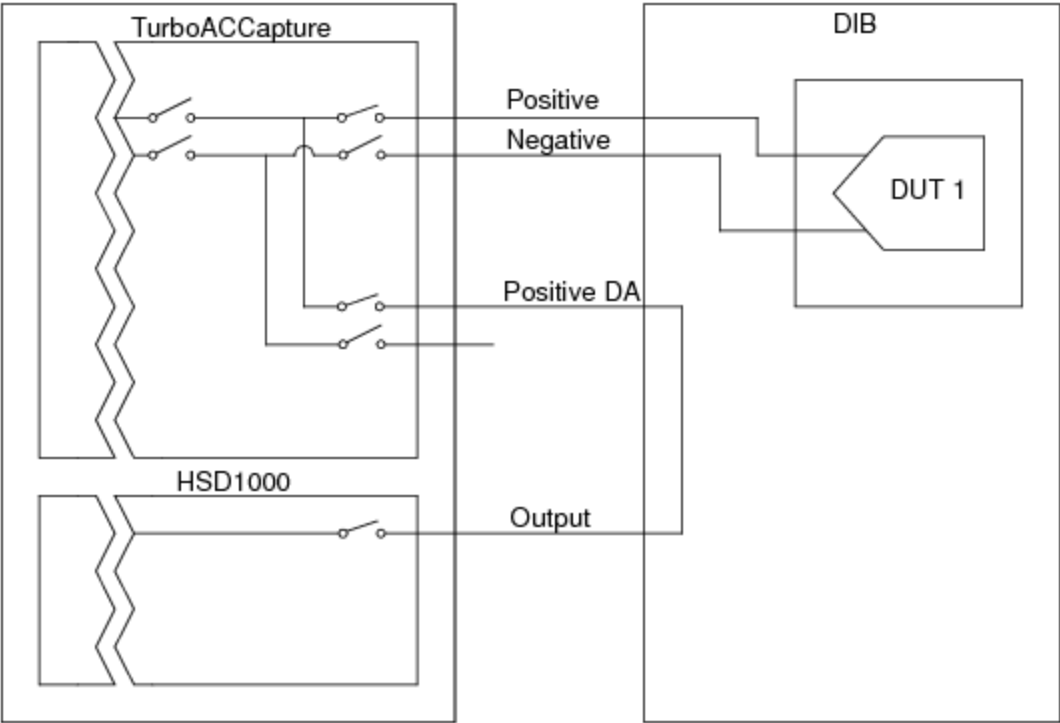
- TurboACCApture as a Primary Resource
- TurboACCApture as Secondary Resource

Note:

Only one DIB Access connection is shared across the four ports. DIB access cannot be used with TurboACSource filters or signal combination.

TurboACCApture as a Primary Resource

When the TurboACCApture is the primary instrument, this means that another instrument pin enters the TurboACCApture DIB access line and connects to the DUT through the TurboACCApture line that is connected to the DUT. The figure shows a diagram of this connection.



TurboAC as Primary Instrument in a DIB Access Connection

The TurboACCapture can only be used as the primary instrument when it is configured in pass through mode or BBAC compatible mode. These connections are described in the following:

- TurboACCapture as Primary Instrument in BBAC Compatible Mode
- TurboACCapture as Primary Instrument in Pass Through Mode

TurboACCapture as Primary Instrument in BBAC Compatible Mode

When the TurboACCapture is in BBAC compatible mode, DIB access connections are programmed similar to BBAC DIB access connections. The following shows a pin map, channel map, VBT syntax, and a diagram of the connections made for an HSD resource delivered as secondary resources to the device pin.

Only a single pin name is required in the pin map. The pin type used is that of the primary instrument, although this is unnecessary. Remember that selecting a certain pin type can change the available pin levels selections.

Pin Map for BBAC Source as a Secondary Resource

Group Name	Pin Name	Type
	OutPos	Analog

The same pin name is entered in the channel map twice, once with the type of BBACCapPos and once with I/O for the digital pin. The :DA syntax indicates that the I/O connection will be made through DIB access.

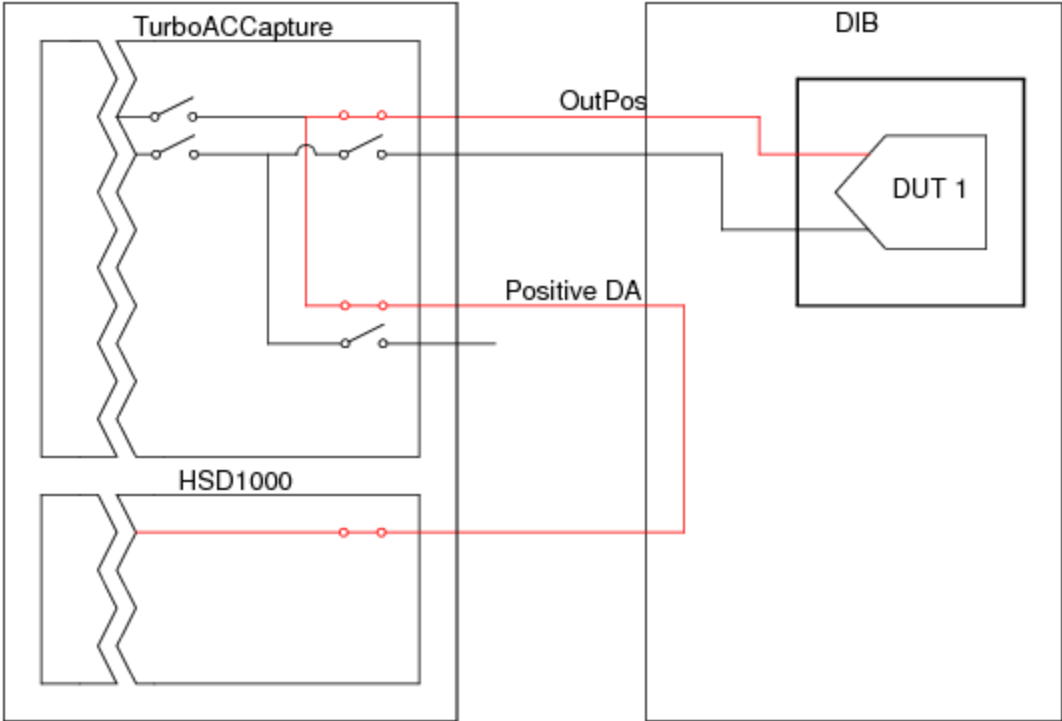
Channel Map for TurboACSource as a Primary Resource

Pin Name	Package Pin	Type	Site 0
OutPos		BBACSrcPos	18.cappos1
OutPos:DA		I/O	1.ch1

Use the following VBT statement to connect to the Digital Pin to the DUT through the TurboACCapture connection:

```
TheHdw.Digital.Pins("InPos").Connect
```

This diagram shows the connection and signal flow that is created.



DIB Access Connection, TurboACSource as a Primary Resource

☒ **Note:**

Proper DIB layout is required when making DIB access connections.

TurboACCapture as Primary Instrument in Pass Through Mode

When the TurboACCapture is in Pass Through mode, you program DIB access connections like typical DIB access connections. The following shows a pin map, channel map, VBT syntax, and a diagram of the connections made for an HSD resource delivered as secondary resources to the device pin.

Only a single Pin name is required in the pin map. The pin type used is that of the primary instrument, although this is unnecessary. Remember that selecting a certain pin type can change the available pin levels selections.

Pin Map for BBAC Source as a Secondary Resource

Group Name	Pin Name	Type
	OutPos	Analog

The same pin name is entered in the channel map twice, once with the type of BBACCapPos and once with I/O for the digital pin. The :DA syntax indicates that the I/O connection will be made through DIB access.

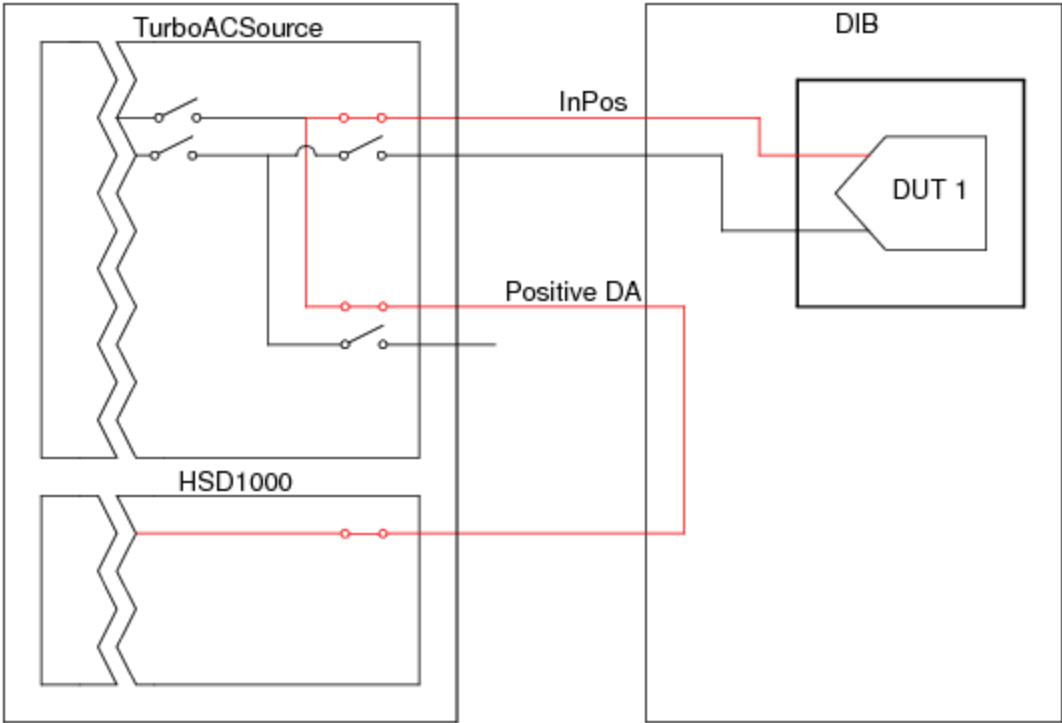
Channel Map for TurboACSource as a Primary Resource

Pin Name	Package Pin	Type	Site 0
InPos		TurboACCapPos	18.cappos1a
InPos:DA		I/O	1.ch1

Use the following VBT statement to connect to the Digital Pin to the DUT through the TurboACSource connection:

```
TheHdw.Digital.Pins("InPos").Connect
```

The diagram shows the connection and signal flow that is created.



DIB Access Connection, TurboACCapture as a Primary Resource in Pass Through

Note:

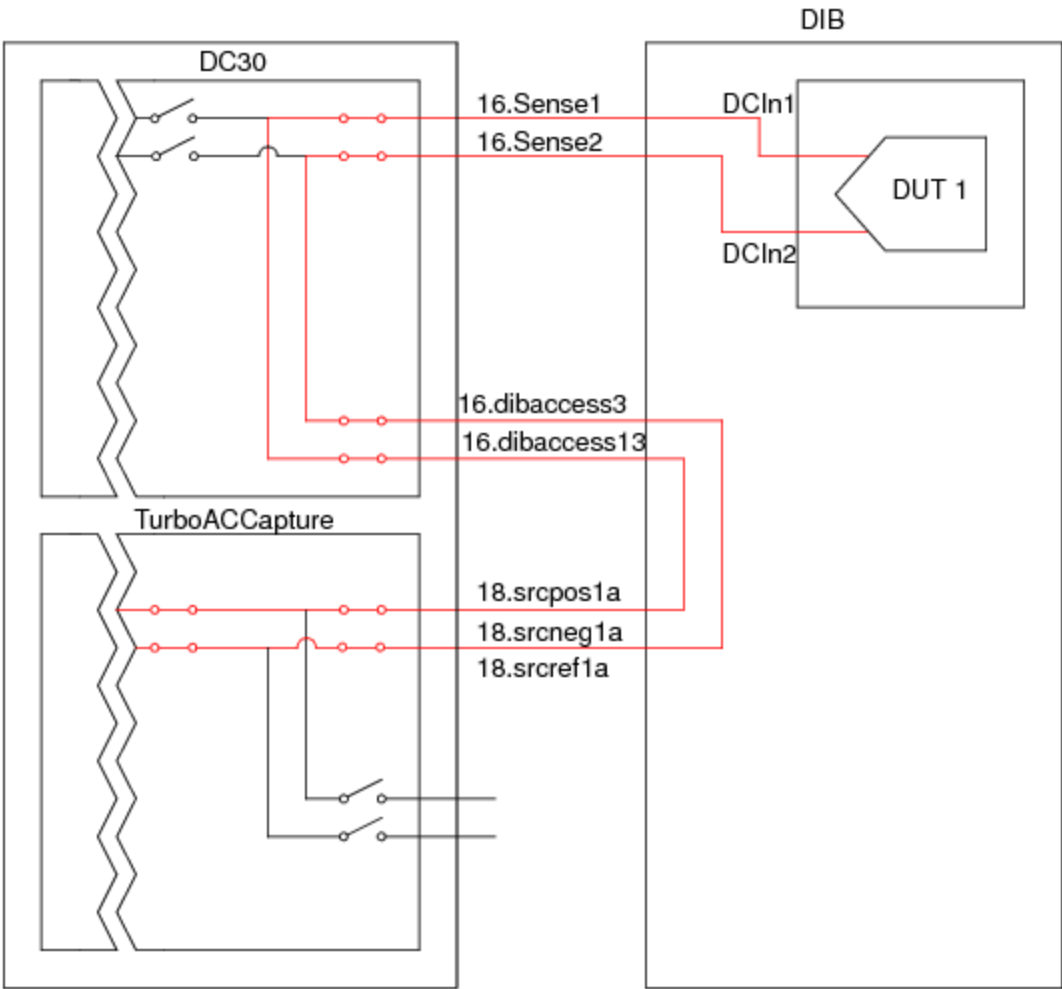
Proper DIB layout is required when making DIB access connections.

TurboACCapture as Secondary Resource

The DIB access use model for the TurboACSource as a secondary resource for a device pin is identical to other UltraFLEX instruments. The following figure shows a TurboACCapture delivered as a secondary resource to the device pins using DIB access.

Channel Map for TurboACSource as a Secondary Resource

Pin Name	Package Pin	Type	Site 0
DC_In1		DCVI	16.sense1
DC_In1:DA		TurboACCapPos	18.cappos1a
DC_In2		DCVI	16.sense2
DC_In2:DA		TurboACCapNeg	18.capneg1a



TurboAC DIB Access, TurboACCapture as a Secondary Resource

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TurboACCApture Programming : Programming Down Conversion

Programming Down Conversion

Down conversion mode allows you to use the TurboACCApture to capture high frequency signals (15MHz to 100MHz) that would normally be beyond the instrument’s direct capture range. For more information, see Down Converter Circuit.

Enabling the Down Conversion Circuitry

Enable the TurboACCApture down converter using the following syntax:

TheHdw.TurboACCApture.Pins("MyPin").DownConversion.Enabled = True

Selecting the Local Oscillator Signal

Select the local oscillator signal using the following syntax:

TheHdw.TurboACCApture.Pins("MyPin").DownConversion.LocalOscillatorSource = tlTurboACCApture

where tlTurboACCApture is one of the following:

- tlTurboACCAptureDIBLO
- tlTurboACCAptureOptionalLO
- tlTurboACCAptureStandardLO
- tlTurboACCAptureUserOptionalLO

Setting Down Conversion Clocking Parameters

The LO frequency is not specified directly by syntax. Instead, values of IF and RF are programmed, and the resulting LO frequency is set according to the equation:

(1) $F_{IF} = F_{LO} - F_{RF}$

1.

To choose these values, start with the device’s fundamental frequency output. This value is F_{RF} .
2.

Decide where in the 15MHz bandwidth of the direct capture circuitry you want the signal to be shifted. This lower frequency value is F_{IF} .

The IF frequency range for TurboACCApture is always baseband (DC to 15MHz). Since the

internal LO synthesizer has a frequency resolution of 1Hz, you must choose values for RF and IF that satisfy the following equation:

(2) $F_{RF} + F_{IF} = \text{an integer}$

3.

Once RF and IF are chosen, you must program the TurboAC’s direct capture’s sample rate (Fs) and number of samples to capture (N) using the coherence equation:

(3) $\frac{F_i}{F_s} = \frac{M}{N}$

Where:

Fi	The input frequency to the direct sampler. For down conversion mode, Fi is FIF.
Fs	The sampling rate of the direct sampler.
N	The bin number corresponding to Fi in the frequency spectrum.
M	Any integer value.

4.

Set the IF and RF frequencies using the following syntax:

```
TheHdw.TurboACCapture.Pins("MyPin1").DownConversion.Frequency = 100000
TheHdw.TurboACCapture.Pins("MyPin1").DownConversion.IntermediateFrequency = 0
```

Note:

If you use the AUX or DIB input to provide the LO to the down converter, then the above syntax is unnecessary and will not affect TurboACCapture. For such cases, set the LO frequency by programming the external instrument’s output frequency to the LO frequency directly.

5.

Set the capture sample rate and the number of samples to capture using the values Fs and N from step 3.

Setting the Down Conversion Input Attenuator

You can attenuate the TurboACCapture’s input signal using an optional 9dB attenuator. Attenuating the input signal improves the quality of distortion measurements with higher amplitude signals above ~4dBm.

Use the following syntax to set the down converter attenuator:

```
TheHdw.TurboACCapture.Pins("MyPin").DownConversion.AttenuatorEnabled = True
```

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[TurboACCArchive Programming : Programming Down Conversion](#) : Programming the Capture Filter

Programming the Capture Filter

You can optionally switch one of two filters into the capture path, a 1kHz notch filter or a 15MHz low-pass filter. Use the following syntax to set the filters:

' Bypass the filters

```
TheHdw.TurboACCArchive.Pins("MyPin").Filter = tlTurboACCArchiveFilterByPass
```

' Set the 1kHz filter to be used

```
TheHdw.TurboACCArchive.Pins("MyPin").Filter = tlTurboACCArchiveFilter1kHzNotch
```

' Set the 15MHz low-pass filter to be used

```
TheHdw.TurboACCArchive.Pins("MyPin").Filter _  
    = tlTurboACCArchiveFilter15MHzLowPass
```

For more information, see [Filter Set](#).

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TurboACCApture Programming : Programming Input Coupling

Programming Input Coupling

The TurboAC instrument board supports two TurboACCApture input coupling paths, DC and AC. DC coupling is the default. AC coupling blocks $\leq 150\text{Vdc}$.

To program the TurboACCApture to use DC input coupling, use the statement:

`TheHdw.TurboACCApture.Pins("MyPin").Coupling = tITurboACCAptureCouplingDC`

To program the TurboACCApture to use AC (capacitive) input coupling, use the statement:

`TheHdw.TurboACCApture.Pins("MyPin").Coupling = tITurboACCAptureCouplingAC`

For more information, see Coupling and Offset.

Starting AC Precharge

TurboACCApture supports fast precharge of the blocking capacitor when the AC coupled input is used. Operation of the precharge function takes about 8ms (to switch the relay), plus the user specified value set as the PreChargeTime parameter.

Note:

Precharge operates only if TurboACCApture is programmed to use AC input coupling.

To avoid capturing varying DC with the TurboACCApture, program the following sequence:

1. Turn off the AC component of your input signal.

2. Start the TurboACCApture with a user specified pre-charge time:

`TheHdw.TurboACCApture.Pins("MyPin").PreCharge 0.002`

3. Turn on the AC component of the input signal.

4. Capture the data.

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TurboACCApture Programming : Programming Voltage Ranging

Programming Voltage Ranging

The TurboACCApture supports voltage ranging.
Use the following statement to set the voltage range:
TheHdw.TurboACCApture.Pins("MyPin").VoltageRange = 2
Any value can be programmed as a voltage range, but the nearest higher range is used.
Programming a range higher than the highest range returns a run-time error.
For more information, see Voltage Ranging.

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TurboACCApture Programming : Starting a TurboACCApture

Starting a TurboACCApture

Start a TurboACCApture using the following VBT syntax:
TheHdw.TurboACCApture.Pins("MyPin").Signals("MySignal").Trigger

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[TurboACCApture Programming](#) : Visual Basic for Test

Visual Basic for Test

[TurboACCApture](#) programs are created using VBT language in the VBT editor.
[For TurboACCApture specific VBT programming](#), see [TurboACCApture](#) in the VBT Syntax Reference.

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TurboACCApture Programming : TurboACCApture Reset Values

TurboACCApture Reset Values

This table lists the local reset values for the TurboACCApture. Pre-job reset, post-job reset, power-up reset, and user reset all result in the same state.

Reset Settings	
Setting	Reset Value
Alarm	All alarms are enabled.
BlockCount	1
SampleSize	256
BlockSpacing	0
ConditionBitEnabled	True
Coupling	tITurboACCAptureCouplingDC
Down conversion attenuator enabled	False
Down conversion enabled	False
Down conversion frequency	20 MHz
Down conversion intermediate frequency	0.0
Down conversion local oscillator source	tITurboACCAptureStandardLO
External trigger load	The loaded command is removed.
Extra samples	0
Filter	tITurboACCAptureFilterBypass

MicroCodeControl	tlTurboACCaptureMicroCodeControlByCPU
Offset auto recharge enabled	True
Offset enable	tlTurboACCaptureOffsetDisable
Offset value	0.0 V
Sample rate	1.0 MHz
Signal cancellation enabled	False
Voltage range	1.28 V

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[TurboACCApture Programming](#) : TurboACCApture Pattern Control

TurboACCApture Pattern Control

You can control a TurboACCApture instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and parameter sets (PSets).

You create a pattern independently of the test program using the PatternTool or a text editor.

Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as defining the pattern microcode used for controlling the instrument during pattern execution.

Note:

To use pattern microcodes to control the TurboACCApture on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

The following topics are available:

- [Programming Patterns in a Text Editor](#)
- [TurboACCApture Pattern Microcodes](#)
- [PSets](#)

Pattern Compiler

The pattern compiler compiles a text .atp file into a binary .pat file. You can edit this file using PatternTool, and it is the format accepted by the pattern loader.

Pattern Reverse Compiler

The pattern reverse compiler compiles a binary .pat file into a text .atp file. You can edit this file using a text editor.

PatternTool

PatternTool allows you to read and edit binary pattern files. Using PatternTool, you can display a compiled pattern file in easily readable, column formats. You can also directly edit a pattern file, instead of having to edit the ASCII source file and recompile it. You can then update the tester with the edited pattern file.

PatternTool is useful during debugging, specifically in debugging pattern files. PatternTool can also display HRAM (History RAM) data, as captured on the tester.

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[TurboACCApture Programming : TurboACCApture Pattern Control](#) : Programming Patterns in a Text Editor

[Programming Patterns in a Text Editor](#)

You can create and edit a pattern using a text editor. This form (.atp files) must be compiled into a .pat file before it can be loaded. The pattern contains microcode to set instrument parameters. Use the PSet microcode to program the instrument to the parameters specified in a PSet.

Creating a pattern in this form requires you to format the file in a specific way. A pattern requires that an instrument statement for each instrument is added and that each vector is written out in a particular way.

The following shows how to create a pattern using a text editor:

- [Instruments Statement](#)
- [Vector Statements](#)

[Instruments Statement](#)

You can only use TurboACCApture microcodes with pins that the instruments statement has specified as TurboACCApture. For example:

```
instruments = {  
    DUT_AOUT:TurboACCApture;  
    DUT_AOUT:BBACCApture;  
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

[Vector Statements](#)

If a vector contains any TurboACCApture related microcode other than nop, that vector is placed in SRM.

The format for specifying a microcode on a particular vector is:
(pin-list : instrument = microcode)
or:
((pin-list1 : instrument , pin-list2 : instrument) = microcode)
where:

pin-list	Is the pin name or pin group.
instrument	Is TurboACCApture.
microcode	Is a TurboACCApture microcode.

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TurboACCApture Programming : TurboACCApture Pattern Control : TurboACCApture Pattern Microcodes

TurboACCApture Pattern Microcodes

A series of extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is captured. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start capturing a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used for issuing commands to a TurboACCApture instrument from a pattern.

TurboACCApture Pattern Microcodes

Microcode	Description
Trig <SignalName>	The instrument immediately starts capturing the specified capture signal. This microcode takes an optional argument, which is the name of the signal to be captured. If the signal name is unspecified, the default signal is used.
Resync	Forces phase alignment of the TurboACCApture instrument clock with the HSD T0 clock.
PSet <PSetName>	Executes the specified parameter set. This microcode requires a single argument, which is the name of the PSet to be applied.
Enable_Inst_Cond	Enables the instrument condition. The TurboACCApture asserts the instrument condition to the pattern generator at the

	completion of a capture.
Disable_Inst_Cond	Disables the TurboACCapture from asserting the instrument condition.
Enable_Alarm	Enables all TurboACCapture alarms.
Disable_Alarm	Disables all TurboACCapture alarms.

Text Editor Example

The following example displays how the TurboACCapture microcode would appear in a pattern file created using a text editor:

```
(MyPin:TurboACCapture = Resync)      > -    ;
```

PatternTool GUI Example

The PatternTool provides columns for the TurboACCapture pins that are listed in the instrument statement of the *.atp file.

The comma-separated microcode equivalent can be conveniently produced by selecting the check boxes as in this figure.



TurboACCapture Microcode Display

☒ **Note:**

TurboAC microcodes have specific rules regarding the necessary spacing in vectors between microcodes on a specific channel.

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TurboACCApture Programming : TurboACCApture Pattern Control : PSets

PSets

PSets have programmable parameters to control the state of an instrument for testing. Each TurboACCApture instrument has parameters that can be programmed using PSets. You define a PSet in VBT language and invoke it using PSet microcode in a pattern or with VBT language.

Calling From Text File

A PSet called from a pattern created in a text file would use the following format:

(MyPin:TurboACCApture = PSet MyPSet)

Calling From PatternTool

A PSet called from a pattern created in PatternTool would look as shown in the following figure.

TSet	Command	Label	Vector/Cy	copipos (TurboACCApture)	Comment
tset1			0		
tset1			1	PSet PSet1	
tset1			2		
tset1			3		

PSet Called From a Pattern Using PatternTool

Calling From VBT

A PSet applied with VBT language would use the following format:

TheHdw.TurboACCApture.Pins("MyPin").PSets.Item("MyPSet").Apply

PSet Example

You define a PSet in VBT whether you are creating patterns with PatternTool or a text editor. The following example shows all PSet programmable parameters being programmed for the TurboACCApture. This PSet can then be invoked from a pattern whenever these parameters are needed.

' Create a PSet and add it to the PSet collection

TheHdw.TurboACCApture.Pins("MyPin").PSets.Add ("MyPSet")

' Program the hardware parameters to the desired values

```
With TheHdw.TurboACCApture.Pins("MyPin").PSets.Item("MyPSet")  
    .BlockSpacing.Value = 0  
    .ExtraSamples.Value = 3  
    .SampleRate.Value = 1000000  
    .SampleSize.Value = 256  
    .VoltageRange.Value = 5
```

End With

' Load the PSet into PSet memory

```
TheHdw.TurboACCApture.Pins("MyPin").PSets.Item("MyPSet").Apply
```

The following table lists the parameters that can be programmed in a PSet for the TurboACCApture and a description of each property.

TurboACCApture PSets	
PSet Property	Description
BlockCount	Sets the number of blocks to be captured. The total capture size is calculated as (SampleSize + ExtraSamples) * BlockCount samples.
BlockSpacing	Sets the number of samples skipped between captured blocks.
ExtraSamples	Sets the extra samples to be programmed.
SampleRate	Sets the sample rate value.
SampleSize	Sets the sample size to be programmed.
VoltageRange	Sets the VoltageRange.

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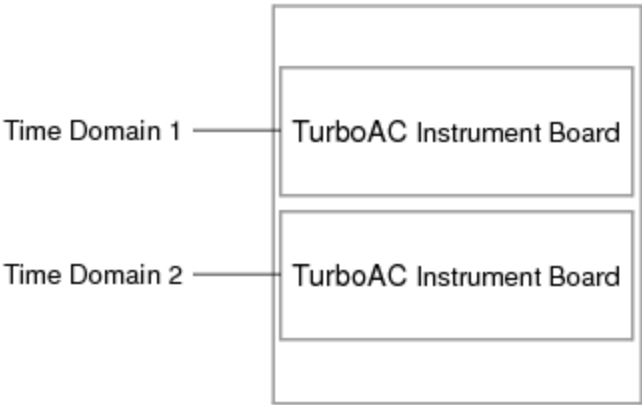
TurboACCapture Programming : Using TurboAC in Concurrent Testing

Using TurboAC in Concurrent Testing

To support concurrent testing, you can assign each TurboAC instrument to a unique time domain. Concurrent features for TurboAC include:

- Each TurboAC board can be configured for one time domain
- Each TurboAC instrument can be controlled by a unique time domain using Sync-Link. Features include:
 - Independent access to Sync-Link
 - Independent sample clock programming, one sample rate per channel
 - Independent clock synchronization with other TurboAC instruments or with digital clocks
 - Independent control by digital pattern
 - Independent capture memory and triggers
- Multiple instruments can be grouped into a common time domain
- TurboAC is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with each TurboAC instrument assigned to its own time domain.



TurboAC Concurrent Testing Configuration Example, 2 Time Domains
For more information, see [About Concurrent Test](#).

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TurboACCapture Theory of Operation

TurboACCapture Theory of Operation

The TurboACCapture instrument set provides AC waveform capture capabilities for testing devices such as precision A/D and D/A converters, high speed A/D and D/A converters, and high performance SOC's such as Set Top Box devices.

Typical tests performed using the TurboAC instruments include gain accuracy, frequency response, distortion, and signal-to-noise ratio.

The TurboACCapture instrument can capture high-accuracy, low-noise, and distortion signals with frequency components of up to 150MHz.

The following topics are available:

- | | |
|---|---|
| • | TurboACCapture Overview |
|---|---|
- | | |
|---|--|
| • | TurboACCapture Signal Path |
|---|--|

The TurboACCapture is a dual source per each TurboAC instrument board. Each board also includes two TurboACSource instruments. For more information, see [About the TurboACSource](#). For information on TurboACCapture instrument performance specifications and standards, see [TurboAC Specifications](#).

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TurboACCapture Theory of Operation : TurboACCapture Overview

TurboACCapture Overview

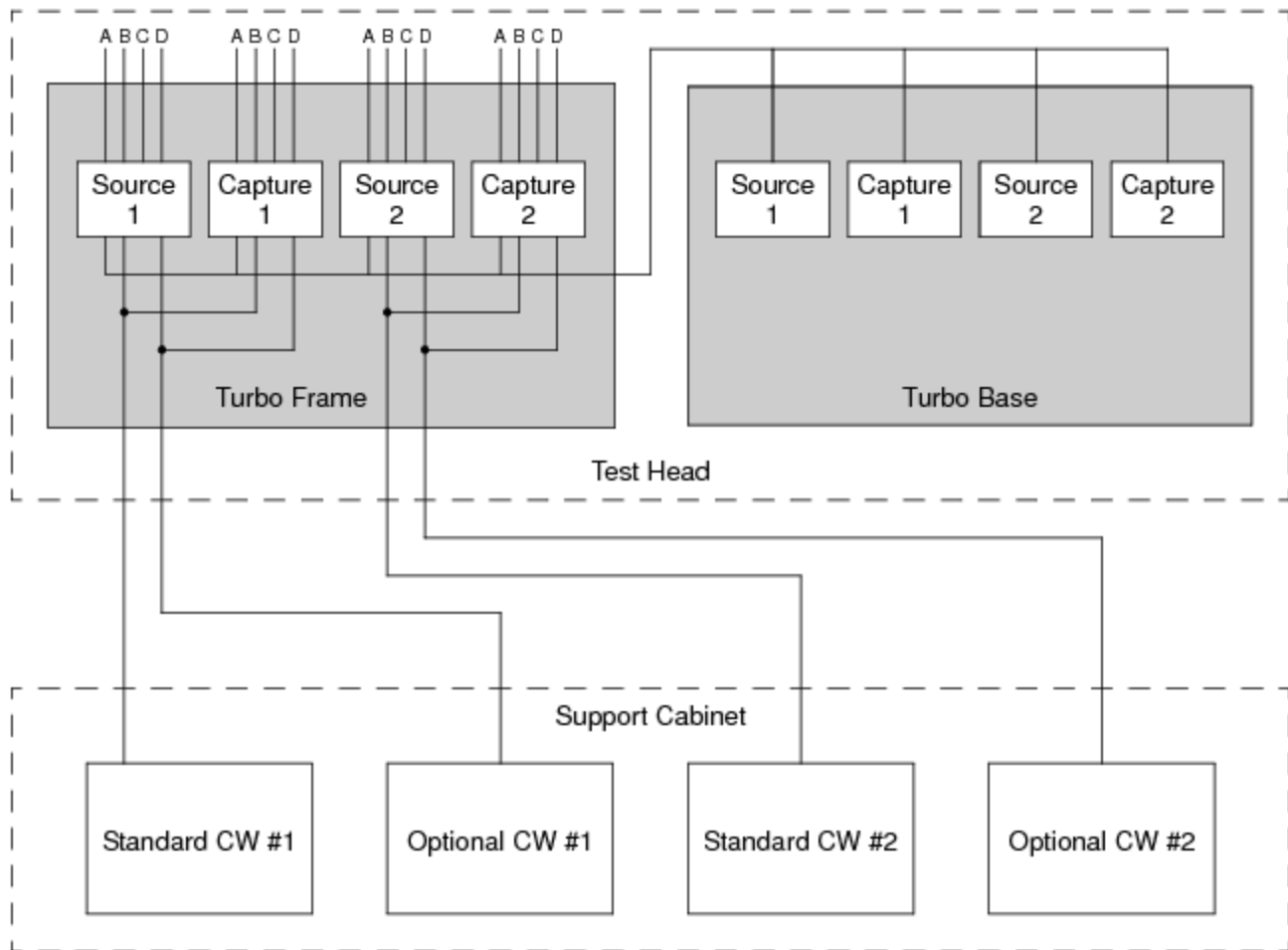
The TurboACCapture instrument can capture high-accuracy, low-noise, and distortion signals with frequency components of up to 15MHz in direct capture mode and up to 100MHz in down conversion mode.

The TurboACCapture has a programmable filter set that includes a 1kHz notch filter with gain and a 15MHz low-pass filter. It also includes a signal cancellation circuit. The TurboACCapture’s input can be switched to one of four differential ports (eight single-ended) by a multiplexed relay tree. You program the TurboACCapture using the VBT programming syntax, and you can also control it using pattern microcodes. Captured data can be processed using DSP. The TurboACCapture also provides an interactive debug display.

The TurboACCapture offers the following modes of operation:

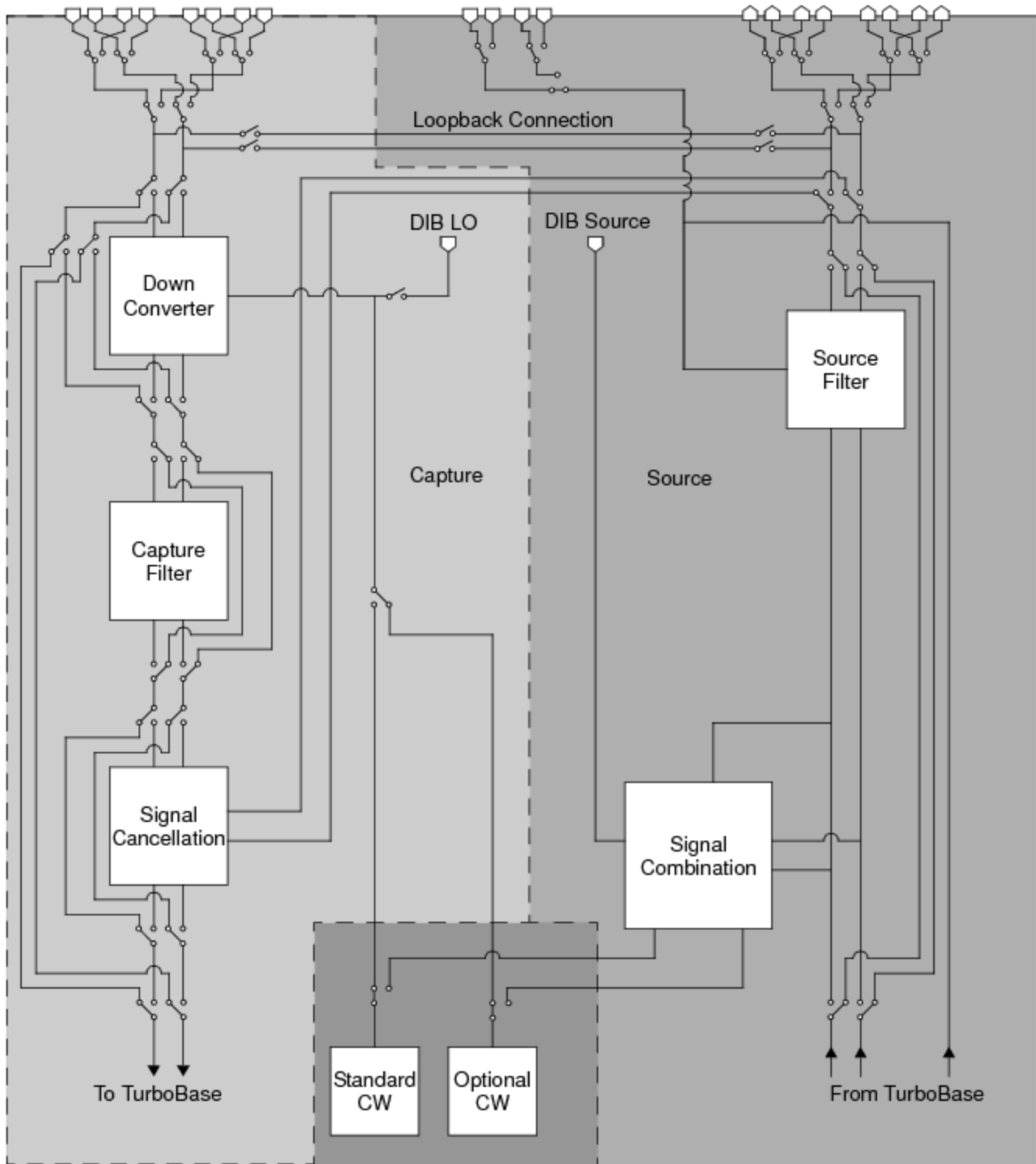
Direct Capture	Direct capture mode allows you to capture and process signals in a bandwidth from DC to 15MHz.
Down Conversion	Down conversion mode allows you to capture and process high frequency signals in the 15MHz to 150MHz bandwidth.
Signal Cancellation	Signal cancellation mode allows you to reduce the dynamic range of the input signal by attenuating the fundamental signal amplitude of the device output.

The TurboACSource hardware set consists of a Turbo Frame Board, a Turbo Base Board, and optional devices from the support cabinet. The following is a functional diagram.



System Level Diagram of TurboAC Hardware

The following is a block diagram of the TurboAC Frame Board.



Single Channel on TurboAC Frame Board

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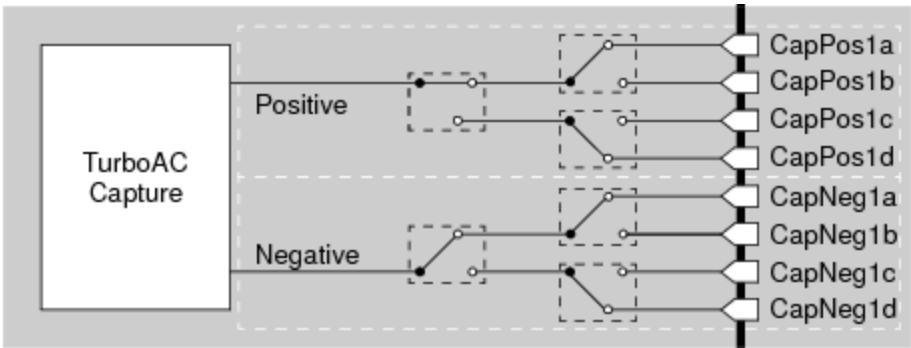
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[TurboACCapture Theory of Operation](#) : [TurboACCapture Overview](#) : TurboACCapture Port

TurboACCapture Port

Each TurboACCapture connects to the DIB through a Port. A Port consists of multiplexed channels allowing each pin to connect to one of four outputs. The figure shows details of the Port.



TurboACCapture Port Connections

When a connection is made to the DUT, all pins of a channel must connect to the same Port output. For example, if the positive pin is connected to the A Port, then the negative and pin must also be connected to the A Port.

If an illegal connection is attempted, an error is generated.

For more information, see [TurboACCapture Connections](#).

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[TurboACCapture Theory of Operation](#) : [TurboACCapture Overview](#) : Internal Loopback Circuitry

Internal Loopback Circuitry

The TurboAC instrument supports internal analog loopback between the source and capture (Source 1 and Capture 1, Source 2 and Capture 2). This feature can be used to verify instrument performance and setup conditions without requiring a DIB or external instrumentation. Each source and capture instrument has an analog loopback relay path devoted to this function. You can source and capture a signal within the same instrument board, which eliminates influences outside of the instruments themselves.

For more information, see [Connecting the Loopback Circuitry](#).

Internal Cross Channel Loopback

Loopback can also be made by crossing to the opposite channel number (Source 1 and Capture 2, Source 2 and Capture 1). This feature is useful when either the source or capture in a standard loopback pairing is unavailable for use.

One use would be sourcing from the CW on source #1 and capturing with down conversion on capture #2, which cannot be done in channel loopback due to the single standard CW restriction. For more information, see [Connecting the Loopback Circuitry](#).

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[TurboACCApture Theory of Operation](#) : [TurboACCApture Overview](#) : CC Common

CC Common

The CC Common is a connection to ground on the channel card, allowing you to supply the voltage reference at the DIB for best noise performance. CC Common can be connected to the DUT, CC ground, or to DGS.

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[TurboACCCapture Theory of Operation](#) : [TurboACCCapture Overview](#) : [Device Ground Sense](#)

Device Ground Sense

Device ground sense (DGS) is a sense signal that tracks the voltage of ground on the DUT and is used as a correction by the instrument to accurately force or measure voltages relative to the ground potential of the device. DGS must be connected to a single point on the DIB. The optimum place to connect DGS is at the device ground pin.

DGS connections are made by IG-XL when the instrument is connected to the DUT. Any used DGS must be grounded on the DIB or a DGS alarm can occur.

 Note:

DGS pins are intended as a ground reference for the PPMUs and not for the TurboACSource instrument. Misapplication of a DGS connection can result in the instrument not realizing its performance specifications.



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TurboACCapture Theory of Operation : TurboACCapture Overview : External Trigger

External Trigger

The TurboAC instrument board features an external trigger pin for each TurboAC instrument, for a total of four pins per board. The external trigger pin is an input pin to the TurboAC instrument that allows you to start sourcing a signal or to trigger a capture based on an external event. You can use the external trigger functionality to synchronize the TurboAC with other instruments in the tester by sending a signal from the DIB to any of the four AC channels on the TurboAC.

Only use the external trigger functionality when you cannot use pattern control for the instrument/DUT synchronization. When using the external trigger, the start or trigger is synchronized to an external rising edge of a pulse from a digital channel that is presented to the External Trigger input of a TurboAC. The following VBT statement sets the signal to be triggered by the external event:

```
TheHdw.TurboACCapture.Pins("MyPin1").ExternalTrigger.Load _
    tlTurboACCaptureExternalTriggerCommandTrigger, "MyCapture"
```

You control the TurboAC’s external trigger pins using VBT language statements that allow you to select microcodes (like START, TRIG) to be executed when an event occurs at the trigger input.

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TurboACCApture Theory of Operation : TurboACCApture Overview : Amplitude Flatness Correction

Amplitude Flatness Correction

To improve the TurboACCApture flatness beyond the instrument’s performance specification, DSP preprocessing can be applied to the capture signal to compensate for the capture instrument’s frequency response. The amplitude flatness correction is applied by the DSP preprocessor by performing the following steps:

1. Perform an FFT of the time domain capture data.
2. Multiply by compensation factors to apply the flatness correction.
3. Perform inverse FFT of the frequency domain data (optional).

Amplitude flatness correction can be applied as follows:

No Correction	Amplitude flatness correction is not applied to the captured signal. Default.
Time Domain Correction	Amplitude flatness correction is applied to the captured signal, and the corrected time domain signal is returned.
Frequency Domain Correction	Amplitude flatness correction is applied to the captured signal, and the corrected spectrum of the signal is returned.

Note:

The frequency domain correction executes faster because an inverse FFT is not required. Because the DSPWave data is in the frequency domain, however, user-DSP code may need to be modified.

[See DSP Example When Using Flatness Correction.](#)

✓ Note:

Amplitude flatness correction is only intended for coherent time domain capture signals. Since incoherent capture signals cause spectral leakage when an FFT is performed, applying the flatness correction can introduce error in the signal. Therefore, using extra samples and/or block captures may not work with capture flatness correction.

✓ Note:

Phase information is preserved when the amplitude flatness correction factor is applied.

Use the following VBT statements to program the correction mode for the capture signal:

```
TheHdw.TurboACCapture.Pins("MyPin").Signals.Add "SignalName"
```

```
TheHdw.TurboACCapture.Pins("MyPin").Signals("SignalName")
```

```
.FlatnessCorrection = Type
```

Where the value of Type can be one of:

- `tlTurboACCaptureFlatnessCorrectionNone`

- `tlTurboACCaptureFlatnessCorrectionTimeDomain`

- `tlTurboACCaptureFlatnessCorrectionFrequencyDomain`

✓ Note:

For amplitude flatness correction, specify the property of the signal before binding the DspWave object to the capture signal.

[DSP Example When Using Flatness Correction](#)

```
Public Function CalcAC_Params(ByVal CapWave As DSPWave, Amplitude As Double, SNR As Double) As Long
```

```
    Dim Spectrum As New DSPWave
```

```
    If CapWave.Domain = TimeDomain Then
```

```
        ' This is the case if amplitude corrections are not enabled or the
```

```
        ' amplitude correction type is time domain
```

```
        Spectrum = CapWave.Spectrum
```

```
    Else
```

```
        ' This is the case if the amplitude correction type is frequency domain
```

```
        Spectrum = CapWave
```

```
    End If
```

```
    Amplitude = Spectrum.CalcAmplitudeFromSpectrum
```

```
    SNR = Spectrum.CalcSNR
```

```
End Function
```

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[TurboACCApture Theory of Operation](#) : [TurboACCApture Overview](#) : Condition Bit Programming

Condition Bit Programming

The TurboACCApture instrument can produce a condition bit when a capture is complete. This allows a pattern to enter a capture loop so that it can keep running (producing DUT clocks, digital I/O, and so on) until the capture completes. When the capture completes, the pattern can either halt or set a flag to indicate to the host computer that the capture is finished. You can then suspend execution of the test program until the capture completes, at which time it is safe to continue on to set up the next test.

This use of the condition bit allows for flexible capture sizes. It is also easier than the error-prone technique of counting pattern vectors and converting into an equivalent time duration to determine which vector will be executing when the digitizer completes its capture. Another common technique is to use a fixed loop size in the pattern, but this does not always work with different capture timing solutions. A fixed loop size can exit too early if the capture is longer than expected, or it can be inefficient and run too long if the capture is shorter than expected.

Each TurboACCApture can have its own microcode column in a pattern. Using the `Enable_Inst_Cond` and `Disable_Inst_Cond` microcodes, you can open and close a window that allows the end of capture instrument condition pulse to move to the pattern generator. This allows you to dynamically select which instrument delivers a condition bit during a pattern burst. Note that if the window is not open at the time when the capture completes, the pulse is missed.

In addition to pattern microcode control of the TurboACCApture instrument condition bit, you can use the following VBT code to control enabling and disabling the bit:

' Enable the capture instrument condition bit:

```
TheHdw.TurboACCApture("MyPin").ConditionBitEnabled = True
```

' Disable the capture instrument condition bit:

```
TheHdw.TruboACCApture("MyPin").ConditionBitEnabled = False
```

All UltraFLEX test system instrument condition bits are OR'ed together before being sent to the pattern generator. Therefore, only one instrument should have its condition bit window enabled at a time to ensure that the specific instrument has completed its capture or reached another state that sets its condition bit.

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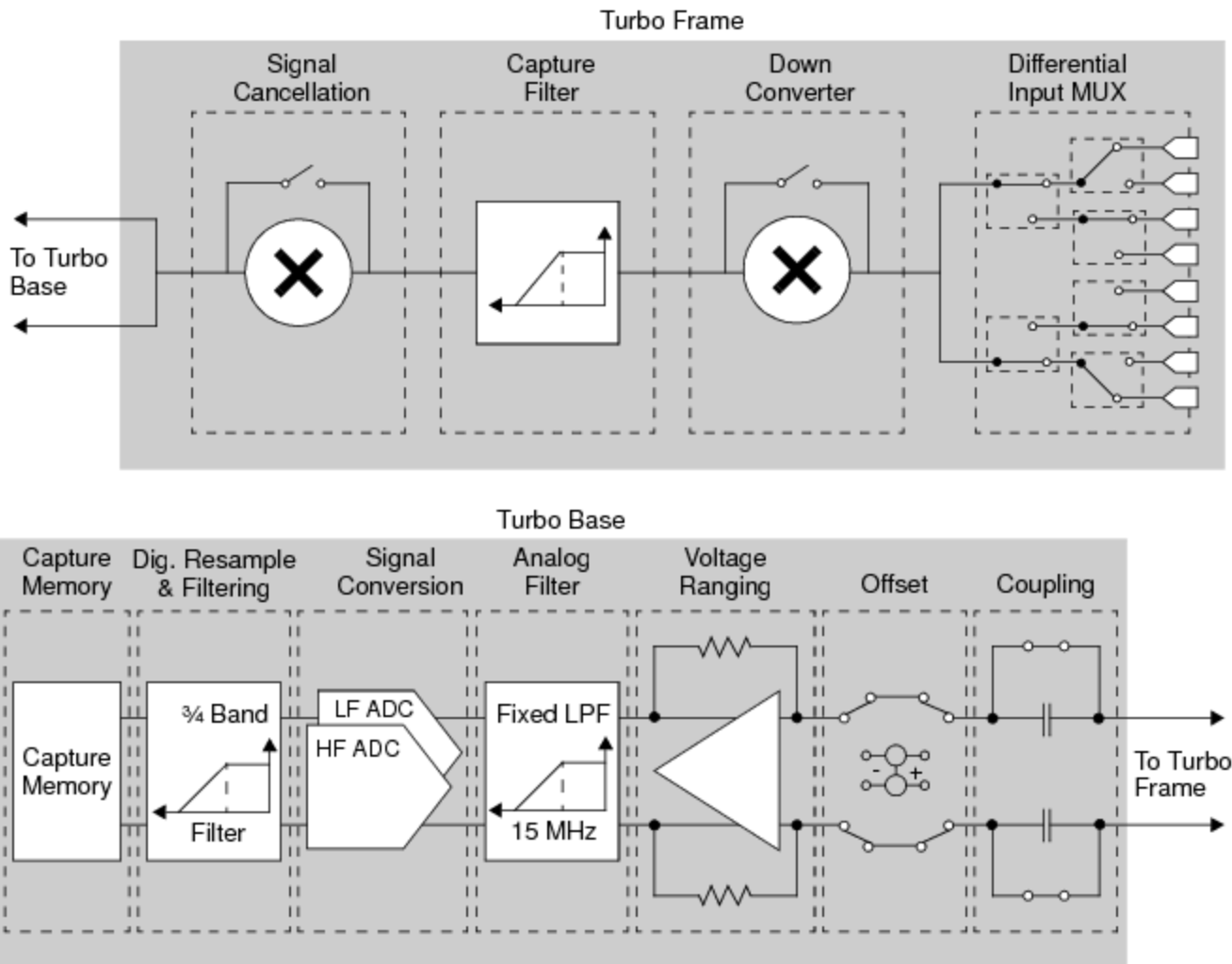
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[TurboACCCapture Theory of Operation](#) : TurboACCCapture Signal Path

TurboACCCapture Signal Path

This figure shows that the TurboACCCapture is a differential instrument that receives analog input signals on its two pins (Positive and Negative) from the DUT. The signal is passed through an optional coupling and offset stage before the signal's amplitude is scaled in the voltage ranging stage. An analog anti-aliasing filter eliminates any high frequency signal components and ensures the TurboACCCapture's target bandwidth of 15MHz. The signal is then sampled at a fixed sample rate of 100MHz using analog-to-digital converters.



TurboACCApture Instrument Components

After the fixed sample rate conversion, the data passes through a digital resampling filter which resamples the capture data at the programmed sample rate. The data is then stored in the instrument's capture memory until the instrument performs an instrument initiated move (IIM). The IIM moves the data to the support board's memory or to the host computer for signal analysis. Details of this process are described in the following topics:

- [Differential Input](#)
- [Down Converter Circuit](#)
- [Filter Set](#)
- [Signal Cancellation](#)
- [Coupling and Offset](#)

- [Voltage Ranging](#)
- [Analog Low-Pass Filter](#)
- [Signal Conversion](#)
- [Digital Resampling and Filtering](#)
- [Capture Memory](#)

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TurboACCApture Theory of Operation : TurboACCApture Signal Path : Differential Input

Differential Input

An input signal enters the TurboACCApture instrument as an analog signal from the DUT. The signal is an arbitrary waveform that must have an amplitude less than the programmed voltage range and frequency components less than the instrument’s bandwidth to be accurately captured. Because the TurboACCApture is a differential instrument, it is always associated with two physical wires.

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TurboACCapture Theory of Operation : TurboACCapture Signal Path : Down Converter Circuit

Down Converter Circuit

The down conversion circuit allows you to use the TurboACCapture to capture high frequency signals (15MHz to 100MHz) that would normally be beyond the instrument's direct capture range. The TurboACCapture's down conversion process begins with attenuation of the input signal by an optional 9dB attenuator, which can be bypassed. Attenuating the input signal allows better distortion measurements with higher amplitude signals above ~4dBm.

The TurboACCapture uses a mixer to shift the higher frequency signal to lower frequencies by an exact known value. The down converter's mixer is a three-port nonlinear device that has two input ports (RF and LO) and one output port (IF). The RF input is the signal entering the instrument from the DUT, and the local oscillator (LO) input signal is generated by either the TurboAC's internal synthesizer, DIB input, or AUX input. The IF output signal of the mixer is the product of the input signal on the two input ports under ideal circumstances. The mixer shifts frequency contents from the device output to a lower frequency, which can be captured by the TurboACCapture's direct capture mode circuitry. However, the mixer also shifts the DUT signal to a higher frequency band, which is not desirable.

The signal out from the mixer is filtered by a 10MHz low-pass image rejection filter, which removes the undesired high frequency content.

The signal is then amplified by a 9dB fixed gain stage. This improves the dynamic range achievable by the direct capture mode circuitry, especially when the 9dB attenuation is used at the front end of the down conversion process. The gain stage cannot be bypassed regardless of the attenuation setting.

TurboACCapture LO

A CW synthesizer specific to the TurboAC instrument is used to create LO signals for the TurboACCapture's down conversion capability. The synthesizer has a frequency range of 1MHz to 310MHz.

The Optional LO is an optional user supplied signal source that connects to the TurboAC instrument board and is programmed using GPIB functionality.

 Note:

The PTS synthesizer is a shared resource that can be switched between the CW channel and the down converter LO. An additional synthesizer can be cabled into the AUX input to enable simultaneous operation of the Downconverter and CW channels.

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TurboACCApture Theory of Operation : TurboACCApture Signal Path : Filter Set

Filter Set

You can optionally switch two filters into the TurboACCApture path: a 15MHz low-pass filter and a 1kHz notch filter. You can also bypass the filter set completely.

The 15MHz low-pass filter reduces out-of-band high frequency content that may saturate the input of the direct capture circuitry. This filter is independent of the 10MHz image rejection filter in the down conversion mode. It need not be enabled in this mode.

The 1kHz notch and the subsequent 30dB gain stage are used for precision audio measurements at 1kHz. The filter reduces the dynamic range of the input signal, allowing better noise and distortion measurements than would be achievable by direct sampling alone.

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TurboACCapture Theory of Operation : TurboACCapture Signal Path : Signal Cancellation

Signal Cancellation

The TurboACSource’s output signal can be routed to the TurboACCapture instrument as an input signal for its capture cancellation circuit. The TurboACCapture’s capture cancellation circuit eliminates a tone from a captured signal by using the TurboACSource’s output signal to cancel out the unwanted tone.

One use of this is in intermodulation distortion testing. With one tone of a dual tone signal removed, any captured IMD products would be from the device under test, and not the capture circuit. Another use is in testing harmonic distortion. Removing the fundamental tone allows the capture instrument to capture the device harmonics without generating any of its own. The capture cancellation circuit works by adding a cancellation tone to the signal under test. This tone is equal in magnitude and 180 degrees out of phase with the tone that is to be removed. The cancellation tone is generated by the TurboACSource and can come from any of the AWG, CW, Aux input, or DIB paths.

In practice, the ability to remove a tone is limited by the instrument’s ability to create a signal of the same magnitude and opposite phase of the tone to be removed. This is easiest to create with the AWG.

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TurboACCApture Theory of Operation : TurboACCApture Signal Path : Coupling and Offset

Coupling and Offset

The coupling and offset circuitry of the TurboACCApture allows you to maximize the dynamic range of the instrument by removing DC offset from the capture signal. By removing the differential offset of the signal you can achieve the optimal dynamic range of the capture range selected. Additionally, you can remove common mode offset from the signal (if necessary) to ensure that the signal on each pin with respect to the TurboAC Channel Card Common (CCC) does not exceed the instrument's compliance.

The captured signal can be DC or AC coupled. When DC coupled, the signal's AC and DC components are passed. The advantage of DC coupling is that the instrument settling time is less because the high-pass filter used for AC coupling is bypassed (see 2-4). When AC coupling is applied, the DC component of the signal is blocked and only the AC components are passed. A precharging circuit is available when using AC coupling, which reduces the settling time associated with the AC coupling high-pass filter. The precharge circuit works to quickly match the DC voltage on either side of the capacitor to minimize settling time.

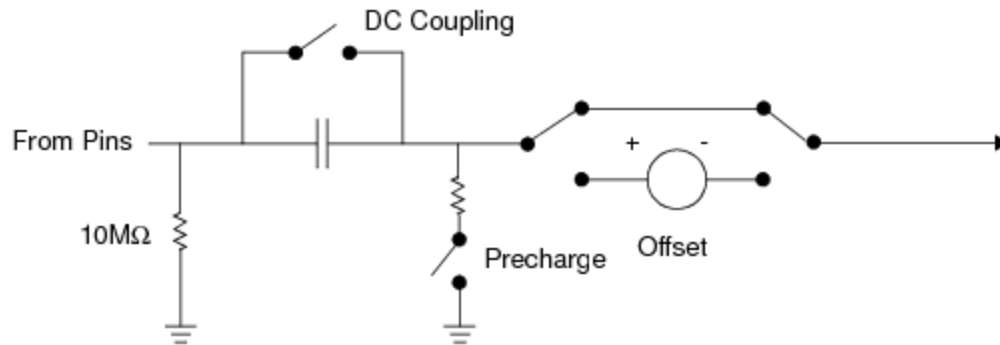
You can apply the DC offset removal circuitry to the positive pin, the negative pin, or both. When the DC offset removal is applied to both pins, common mode offsets can be removed.

Precharge and AC Input Coupling

Use the precharge function on the TurboACCApture instrument to minimize settling time associated with using AC Coupling. The TurboACCApture has a high input impedance which allows the AC coupling capacitor to be small and the low frequency corner to be low. However, this also lengthens the settling time to charge to 2ppm of final DC value.

To reduce settling time, precharge circuits are incorporated in the AC coupling paths. The precharge function works by connecting a low value resistor in parallel with the normal input resistor, allowing the coupling capacitor to rapidly charge to the DC signal component. After the capacitor has charged, the AC signal can be turned on and captured.

The figure shows that the precharge circuit provides a low impedance path to ground, which allows the AC coupling capacitor to charge quickly and reduce settling time.



Model of the Precharge Circuit with AC Coupling

Precharge is specified as a duration (in seconds), and the precharge circuit is activated immediately and remains active for the duration programmed. It is then automatically deactivated. Note that a small transient is associated with the precharge, which must settle. For small voltage steps, the settling time improvement from the precharge is negated by the transient settling time.

You can program precharge using a VBT language statement. See [Starting AC Precharge](#).

☑ **Note:**

The same precharge circuitry is also used by the baseline removal circuitry to quickly reach the programmed DC level. When setting the voltage offset, the precharge is set automatically, but it can be disabled using VBT language.

☑ **Note:**

Only use the precharge feature when the hardware is programmed for AC coupling mode.

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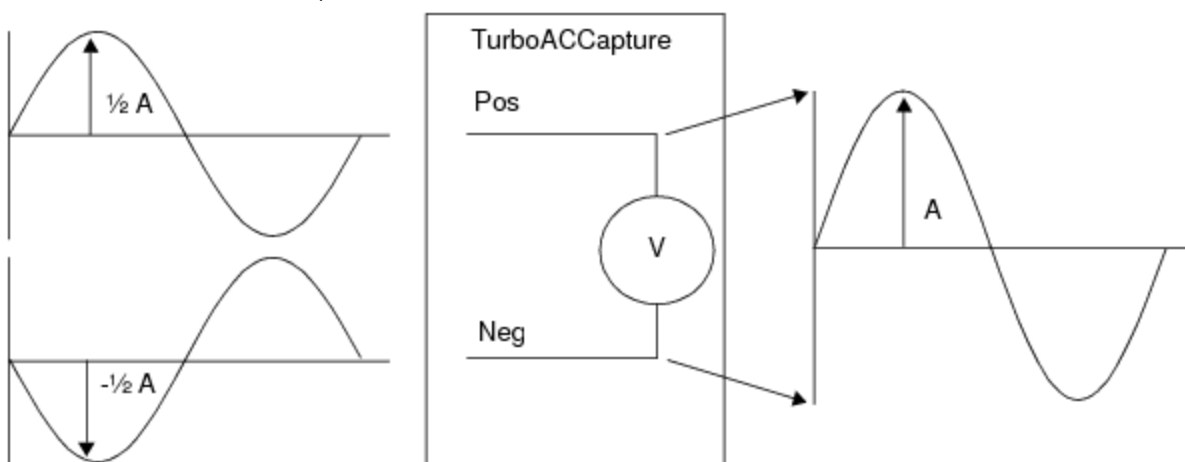
TurboACCApture Theory of Operation : TurboACCApture Signal Path : Voltage Ranging

Voltage Ranging

The range specifies the maximum or minimum voltage that can be applied between the positive and negative pins, or differential voltage. For example, if the voltage between the positive and negative pins is 1.28V, the range must be set to 1.28V or higher to accommodate the signal. Setting the capture to a range smaller than the maximum voltage causes the capture to clip the signal, resulting in an overrange alarm.

Another limitation on the capture is that the voltage on the positive or negative pins must be less than the instrument compliance ($\sim 7V$ with respect to the channel card common).

The 7V constraint applies to the single-ended voltage between each pin and the channel card common, whereas the range applies to the differential voltage between the positive and negative. If the instrument is connected to a balanced differential signal source of 10.24Vpk amplitude (as shown in 2-6, $A = 10.24V$), the positive and negative pins will never exceed 5.12V with respect to channel card common, which is smaller than the 7V limit.



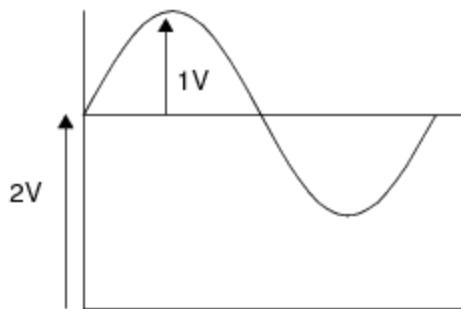
Differential Signal Capture

Valid TurboACCApture ranges are: 10.24V, 7.64V, 5.12V, 3.62V, 2.56V, 1.81V, 1.28V, 0.905V, 0.64V, 0.452V, 0.32V, 0.226V, 0.16V, and 0.113V.

As shown in the figure, the input signal is a balanced differential sine wave with amplitude A . The maximum or minimum instantaneous voltage ever presented between the positive and negative pin is at the peaks, A volts. The capture range must therefore be set to a value greater than or equal to A . This is not the same as the peak-to-peak voltage of the sine wave, which is $2 \cdot A$. Peak-to-peak refers to the maximum voltage difference encountered over time.

When dealing with ranging issues, the instrument is only concerned with the instantaneous voltage. The maximum V_{pp} of a captured sine wave is only twice the range voltage when the signal has no differential offset.

For example, to capture the differential signal shown in the following figure, the instrument should be set to a range of at least $3.62V$ to avoid saturating the capture (assuming the baseline circuit is not used). The $1.28V$ and $2.56V$ ranges would be too small for this signal because the combination of the AC and DC components creates a signal that has a maximum instantaneous voltage differential of $3V$.



Signal with 2V DC Offset

As a final example, consider a loopback between a TurboACSource and TurboACCapture, which sources and captures a signal of $2V_{pp}$, using no offset and the best possible range. The $2V_{pp}$ signal corresponds to a $1V$ amplitude signal, so the source range must be set to the $1.28V$ range to optimally generate this signal. Since the maximum differential voltage generated by the source is $1V$, the capture range must also be set to $1.28V$.

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TurboACCApture Theory of Operation : TurboACCApture Signal Path : Analog Low-Pass Filter

Analog Low-Pass Filter

The scaled input signal passes through a fixed low-pass filter to eliminate any high frequency components. Attenuating frequency components beyond 15MHz minimizes the effects of aliasing from the sampling process of the instrument’s ADCs. This ensures that the instrument meets its performance specifications.

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[TurboACCApture Theory of Operation](#) : [TurboACCApture Signal Path](#) : [Signal Conversion](#)

Signal Conversion

The filtered signal is passed to the analog-to-digital converters (ADCs), which effectively sample the analog waveform at a frequency of 100 Msps. The converted analog signals are separated into high (~3 to 15MHz) and low (~DC to 3MHz) frequency components and are passed on to the digital resampling and filter section for recombination.

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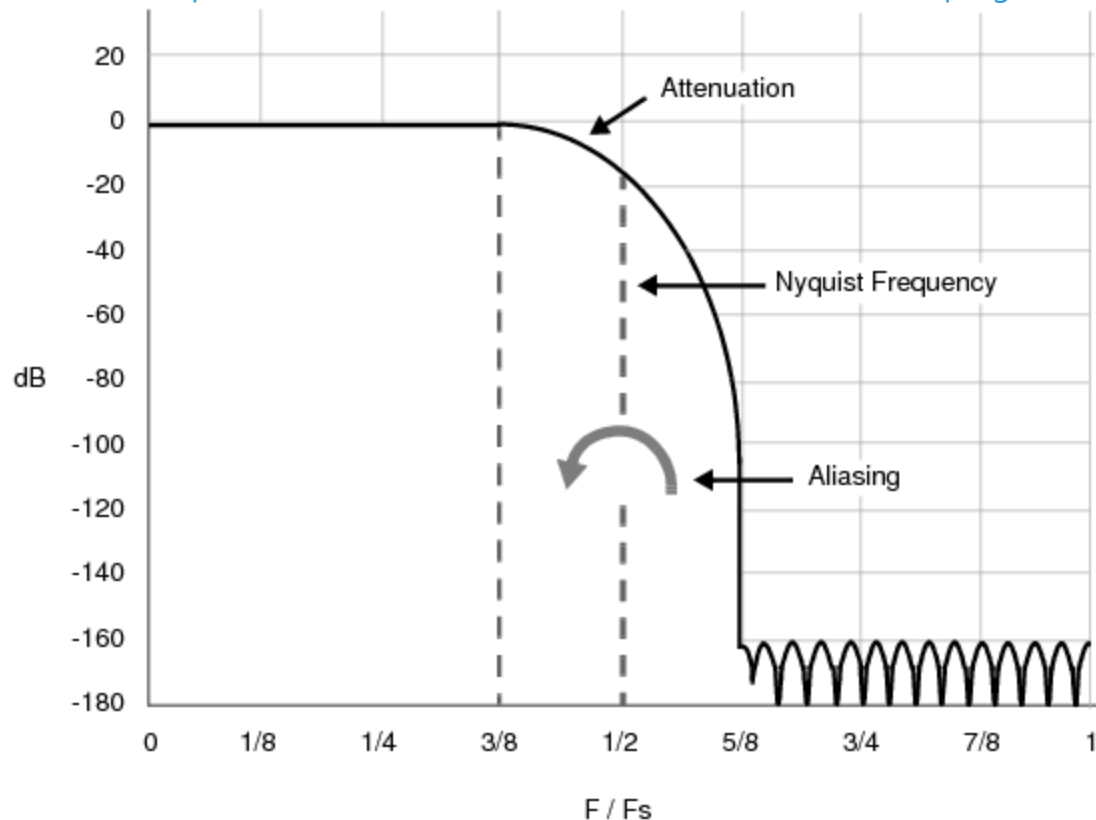
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TurboACCApTure Theory of Operation : TurboACCApTure Signal Path : Digital Resampling and Filtering

Digital Resampling and Filtering

The high and low frequency digital signals are recombined into one signal by a digital crossover filter. This fixed data rate signal is then resampled at the programmed sample rate through stages of interpolation and decimation. The signal is then passed through a 3/4 band filter, which removes any aliasing that was introduced by the resampling process.

The impact of the 3/4 band filter on the capture signal is important. The following figure shows that the frequency response shape of the 3/4 band filter is always the same, while the bandwidth varies with the sample rate. The bandwidth is $0.75 * F_s/2$, where F_s is the programmed sampling rate.



3/4 Band Filter

The spectrum shown in the figure is between $3/8 * F_s$ and $F_s/2$ should not be included in any calculations because:

- The filter has started to roll off, so the frequency response is not flat.
- Any signals or noise present in the range of $F_s/2$ to $5/8 * F_s$ are only partially attenuated and will alias back into this part of the spectrum.

To avoid aliasing, choose F_s using the following procedure:

1. Let BW = highest frequency of interest for the test.
2. Choose an F_s such that $F_s \geq 8/3 * BW$.
3. Restrict all spectral analysis (for example, SNR) to the region of DC to BW .

Note that Mixed Signal Workshop automatically enforces step 2. You need only provide the required bandwidth to the Solver.

		
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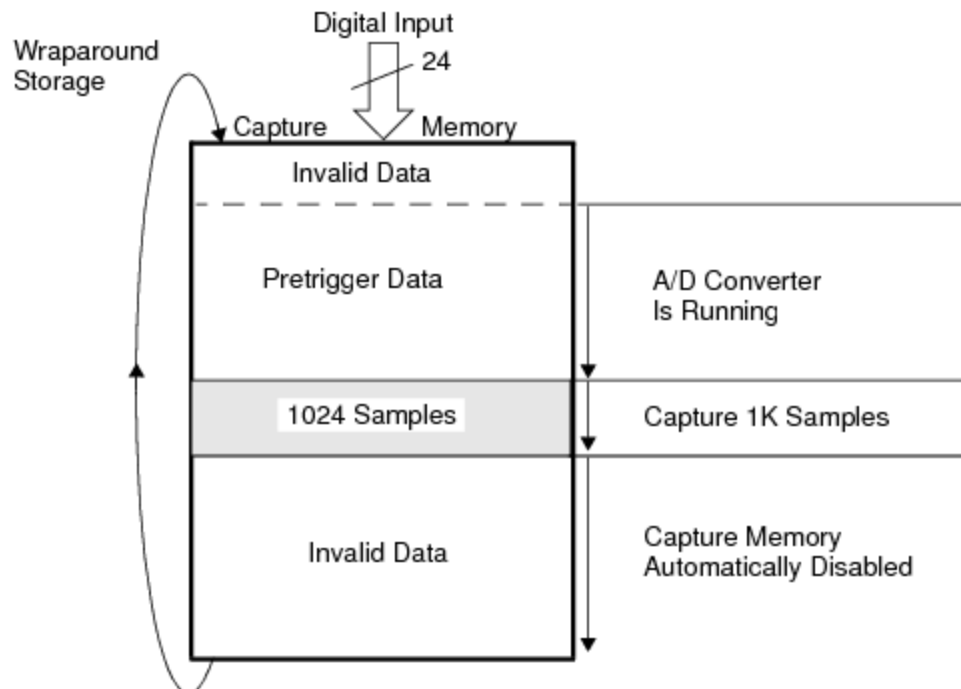


[TurboACCapture Theory of Operation](#) : [TurboACCapture Signal Path](#) : [Capture Memory](#)

Capture Memory

Digital values from the 3/4 band output filter are routed to a 1,048,064-sample capture memory. The TurboACCapture uses an instrument initiated move (IIM) to automatically move the captured data from the instrument's capture memory to the memory in either a system DSP computer or the host computer for signal analysis. Each capture begins where the previous capture ended. As soon as a capture completes, the next capture can begin.

The figure shows an example of this process.



Operation of Capture Memory

Capture Sample Size Parameters

You can program TurboACCapture sample size for a single capture or for a block capture. When performing a single capture, use the `SampleSize` parameter to specify the number of samples to

capture. In addition to SampleSize, capture extra samples using the ExtraSamples parameter, which can be a useful technique to manage device settling times.

The total capture size is calculated as:

$$\text{Total Capture Size} = \text{SampleSize} + \text{ExtraSamples}$$

The following syntax sets the TurboACCapture to capture 256 samples at a rate of 5.000kHz:

```
TheHdw.TurboACCapture.Pins("MyPin").SampleRate.Value = 5000
```

```
TheHdw.TurboACCapture.Pins("MyPin").SampleSize.Value = 256
```

Note:

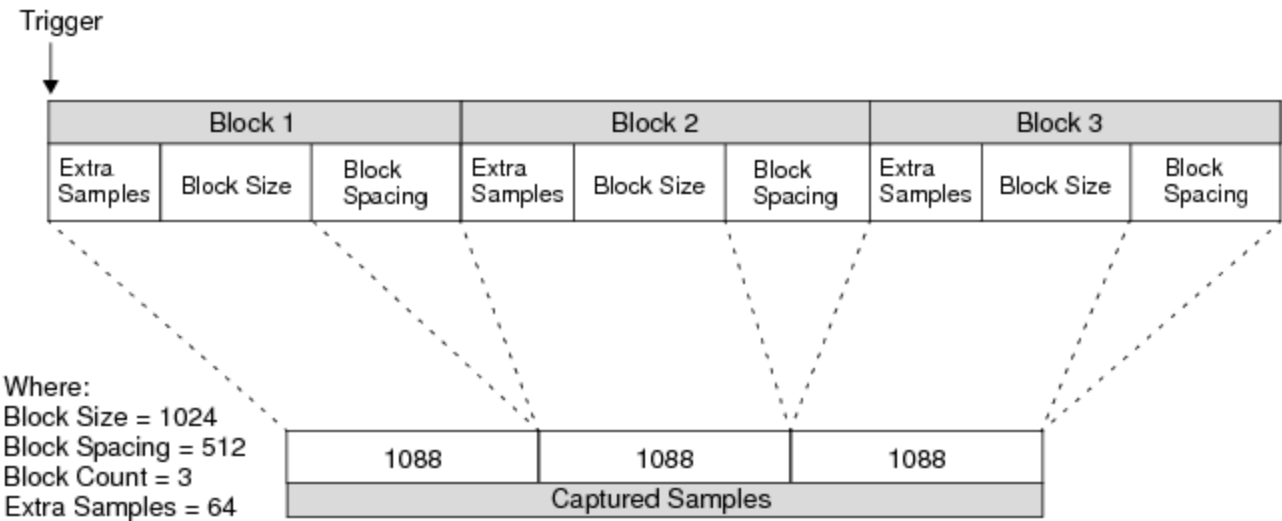
The single capture’s Total Capture Size equation assumes that BlockCount equals 1, which is the default value.

When performing a block capture, the BlockSize and BlockSpacing parameters work with the BlockCount parameter to capture specified sets of samples. The BlockSize parameter specifies the number of samples in each of the one or more blocks you want to capture. The BlockSpacing parameter specifies how many samples are to be skipped (not captured in memory) between each pair of blocks captured. The BlockCount parameter specifies the number of blocks to capture. Additionally, each block can capture extra samples as specified by the ExtraSamples parameter.

The total capture size is calculated as:

$$\text{Total Capture Size} = (\text{BlockSize} + \text{ExtraSamples}) * \text{BlockCount samples}$$

This figure shows an example of using the block capture mode to capture three blocks of data.



Capture Sample Size Parameters

VBT example:

```
TheHdw.TurboACCapture.Pins("MyPin").SampleSize.Value = 1024
```

```
TheHdw.TurboACCapture.Pins("MyPin").BlockSpacing.Value = 512
```

```
TheHdw.TurboACCapture.Pins("MyPin").BlockCount.Value = 3
```

```
TheHdw.TurboACCapture.Pins("MyPin").ExtraSamples.Value = 64
```

This syntax will cause the capture instrument to:

- Capture the first 1088 samples (SampleSize + ExtraSamples)
- Skip the next 512 samples (BlockSpacing)
- Capture the next 1088 samples
- Skip the next 512 samples

and so on, until three blocks (BlockCount = 3) are captured. The capture then stops and the samples can be moved out of capture memory.

 Note:

The BlockSize parameter is an alias of the SampleSize parameter. The SampleSize syntax is more commonly used when block capture mode is not required.





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